

STAT 208: Statistical learning theory

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Probabilistic framework

- Observe feature $X \in \mathcal{X}$ and predict outcome $Y \in \mathcal{Y}$ based on it.
- Of course, there should be some relation between X and Y .
- Examples¹

X	Y
document	topic
image of handwritten digit	digit
image of an object	type of object
image of signature	identity of writer
email	spam or not
sentence	parse tree
medical tests	disease
gene expression in tissue	cancerous or not
phylogenetic profile of a gene	gene function
web search query	ranked list of pages
friendship networks	communities

¹Most from P. Bartlett's slides.

- Model the relation as a joint probability distribution $\mathbb{P}$ on $\mathcal{X} \times \mathcal{Y}$:

$$(X, Y) \sim \mathbb{P}$$

- Often want to learn the conditional distribution $\mathbb{P}_x^* := \mathbb{P}(Y \in \cdot \mid X = x)$ or some aspect of it, e.g. $f^*(x) := \mathbb{E}[Y \mid X = x]$.
- More generally, learn a map $\mathbb{Q} : \mathcal{X} \rightarrow \mathcal{M}(\mathcal{Y})$ where $\mathcal{M}(\mathcal{Y})$ is the set of probability distributions on $\mathcal{Y}$. Useful if we want to associate uncertainty to our predictions.
- Often we settle for learning a function $f : \mathcal{X} \rightarrow \mathcal{Y}$.
- Learning functions is easier than learning distributions. (Function spaces behave like infinite-dimensional vector spaces; easy to apply linear algebraic techniques.)
- The less we can assume about $\mathbb{P}$ the better.
- Parametric models assume $\mathbb{P} \in \{\mathbb{P}_\theta : \theta \in \Theta\}$ for some $\Theta \subset \mathbb{R}^d$.
- Often interested in nonparametric setups.

Risk minimization

- What function are we seeking?
- Pick a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$.
- $\ell(f(X), Y)$ measure how good we are doing on observation (X, Y) .
- Our goal is to recover (or approximate) the function minimizing the risk:

$$f_{\mathcal{F}}^* \in \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{E}[\ell(f(X), Y)]$$

- Here $\mathcal{F}$ could be a pre-specified class of functions encoding our desiderata about the prediction function f (e.g., being Holder continuous of certain degree, etc.)
- Computing $f_{\mathcal{F}}^*$ requires the unknown distribution $\mathbb{P}$.
- Instead, we assume that we observe a *training sample* (key difference with unsupervised approaches)

$$(X_i, Y_i) \stackrel{\text{iid}}{\sim} \mathbb{P}, \quad i = 1, \dots, n$$

Let $D_n := \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ collect our training data.

- Instead, we assume that we observe a *training sample* (key difference with unsupervised approaches)

$$(X_i, Y_i) \stackrel{\text{iid}}{\sim} \mathbb{P}, \quad i = 1, \dots, n$$

Let $D_n := \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ collect our training data.

- We also assume $(X, Y) \sim \mathbb{P}$ is independent of D_n .
- (X, Y) represent a future sample point (*testing sample*) where we observe only the X coordinate and want to predict its Y coordinate.

- Our goal is to recover (or approximate) the function minimizing the risk:

$$f_{\mathcal{F}}^* \in \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{E}[\ell(f(X), Y)]$$

- Given the training data $(X_i, Y_i) \stackrel{\text{iid}}{\sim} \mathbb{P}$, $i = 1, \dots, n$, the *empirical distribution*

$$\mathbb{P}_n := \frac{1}{n} \sum_{i=1}^n \delta_{(X_i, Y_i)}$$

approximates $\mathbb{P}$. In other words, $\mathbb{P}_n$ is a (consistent) estimate of $\mathbb{P}$.

- Replace expectation w.r.t. unknown $\mathbb{P}$ with that w.r.t. known $\mathbb{P}_n$:

$$\hat{f}_n \in \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{P}_n[\ell(f(X), Y)] = \operatorname{argmin}_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \ell(f(X_i), Y_i)$$

(Here we are assuming $(X, Y) \sim \mathbb{P}_n$ when computing the expectation.)

- This is the empirical risk minimization (ERM) idea.

- Target versus the estimate:

$$f_{\mathcal{F}}^* \in \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{E}[\ell(f(X), Y)]$$

$$\hat{f}_n \in \operatorname{argmin}_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \ell(f(X_i), Y_i)$$

- $\hat{f}_n$ can in principal be computed.
- Computation might be intractable due to the complexity of the loss function ℓ or $\mathcal{F}$. $\implies$ try to relax the loss
- $\mathcal{F}$ might be too large for a sample of size n . (Overfitting.) $\implies$ Try to regularize. E.g., optimize over simpler classes $\mathcal{F}_n$ often with the property that $\mathcal{F}_n \uparrow \mathcal{F}$ as $n \rightarrow \infty$.

Optimal (Bayes) rule

- Assuming that $\mathcal{F} = \{\text{All meas. functions from } \mathcal{X} \text{ to } \mathcal{Y}\}$,
- the optimal rule is obtained as the minimizer of the posterior risk:

$$f^*(x) = \operatorname{argmin}_{a \in \mathcal{Y}} \mathbb{E}[\ell(a, Y) \mid X = x]$$

- Show this as an exercise.

Example 1 (Regression)

Assume $\mathcal{Y} = \mathbb{R}$, and consider quadratic loss $\ell(a, y) = \frac{1}{2}(a - y)^2$.

- Optimal (unconstrained) Bayes rule is $f^*(x) = \mathbb{E}[Y \mid X = x]$.
- ERM is obtained as

$$\hat{f}_n \in \operatorname{argmin}_{f \in \mathcal{F}_n} \frac{1}{2n} \sum_{i=1}^n (Y_i - f(X_i))^2$$

- Empirical loss is convex in f .
- Without $\mathcal{F}_n$ restriction, many f achieve zero empirical risk (overfitting).
- Assume $\mathcal{X} \subset \mathbb{R}^d$, and consider

$$\mathcal{F}_n = \mathcal{F}_{\text{lin}} := \{f_{\theta, \theta_0} : \theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}\}$$

where $f_{\theta, \theta_0}(x) = \theta^T x + \theta_0$.

- ERM is usual least-squares regression (unique if design is full-rank.)

Example 2 (Classification)

Assume $\mathcal{Y} = [K] := \{1, 2, \dots, K\}$ and consider zero-one loss $\ell(a, y) = 1\{a \neq y\}$.

- Optimal (unconstrained) Bayes rule is: $f^* : \mathcal{X} \rightarrow [K]$,

$$f^*(x) = \operatorname{argmax}_{k \in [K]} \mathbb{P}(Y = k \mid X = x)$$

that is the maximum a posteriori (MAP) rule.

- The ERM is obtained as

$$\hat{f} \in \operatorname{argmin}_{f \in \mathcal{F}_n} \frac{1}{n} \sum_{i=1}^n 1\{Y_i \neq f(X_i)\}$$

- Here the problem is intractable due to combinatorial nature of loss, even if we drop the constraint $\mathcal{F}_n$, or have a simple $\mathcal{F}_n$.

Choice of the function class

- Linear:

- Notation $x \mapsto \theta^T x$ is the function $f(x) = \theta^T x$.

$$\mathcal{F}_{\text{lin}}(\Theta) := \{x \mapsto \theta^T x : \theta \in \Theta\}, \quad \Theta \subset \mathbb{R}^d$$

- Often assume that the intercept is included for simplicity, i.e.,

$$x = (\tilde{x}, 1), \quad \theta = (\tilde{\theta}, \theta_0), \quad \theta^T x = \tilde{\theta}^T \tilde{x} + \theta_0$$

- Class of affine functions can be considered a linear class in augmented feature vector.

- (Feed-forward) neural networks:
 - A single neuron $\eta(x; \theta, \phi) = \phi(\theta^T x)$.
 - Often multiple layers cascaded together:

$$a_j^{[\ell]}(x) = \eta(a^{[\ell-1]}(x); \theta_j^{[\ell]}, \phi_j^{[\ell]}), \quad j \in [N_\ell], \ell \in [L]$$

- $a_j^{[\ell]}(x)$ output (or activation) of neuron j in layer ℓ .
- $a^{[\ell]}(x) = (a_1^{[\ell]}(x), \dots, a_{N_\ell}^{[\ell]}(x))$ the vector of activations at layer ℓ .
- Layer 0: the input $a^{[0]}(x) = x$ and $N_0 = d$.
- Nonlinear parametric function families: say $N_L = 1$ and $f(x; \theta) = a_1^{[L]}(x)$.
- Often ϕ_j are fixed nonlinearities: ReLU, sigmoid, tanh, etc.
- All parameters: $\theta = (\theta_j^{[\ell]}, j \in [N_\ell], \ell \in [L])$.
- In a fully connected network: $\theta_j^{[\ell]} \in \mathbb{R}^{N_{\ell-1}+1}$.
- Total # of parameters: $\sum_{\ell=1}^L N_\ell(N_{\ell-1} + 1)$.
- Often too many parameters; need to introduce regularization (e.g. sparsity).
- E.g. if every layer has the same # of neurons as the input layer: $\sim Ld^2$.

- Functions spaces as infinite-dimensional vector spaces:
 - We can do basis expansions to represent functions.
 - Let $\psi_1, \psi_2, \dots$ be a basis for $\mathcal{F}$.
 - Any $f \in \mathcal{F}$ can be written as $f = \sum_{j=1}^{\infty} \theta_j \psi_j$ where $\theta_j \in \mathbb{R}$.
 - We can consider “truncated space”

$$\mathcal{F}_N = \left\{ f : f = \sum_{j=1}^N \theta_j \psi_j, \theta_j \in \mathbb{R} \right\}$$

- Can let $N = N_n$ grow with the sample size.
- If $\mathcal{F}$ is a Hilbert space we can take the basis to be orthonormal (many choices).
- E.g., $\mathcal{F} = L^2([0, 1]; \text{Unif})$ with trigonometric basis (Fourier expansion).
- E.g., $\mathcal{F} = L^2(\mathbb{R}; \text{Gauss})$ with Hermite polynomials.

- Reproducing kernel Hilbert spaces (RKHS) of functions:
 - We will talk about these at length.
 - Very natural for machine learning applications where we observe discrete samples of a function.
 - Related to the “kernel trick”.
- Ensemble approaches:
 - Writing a function as a weighted average of weak learners;
 - Some stochastic element (subsampling data) can be introduced in fitting the weak learners (e.g. random forests).
- Sieves: Finite families of functions $\mathcal{F}_N = \{f_1, \dots, f_N\}$.

General theoretical goal

- Let $R(f) = \mathbb{E} \ell(f(X), Y)$.
- Excess risk a measure of prediction performance:

$$\mathcal{E}(f) = R(f) - R(f^*) = R(f) - \min R(\cdot)$$

- Ideally, we would like to show $\mathcal{E}(\hat{f}_n) \rightarrow 0$ as $n \rightarrow \infty$. (Prediction consistency).
- Ideally, characterize how fast it goes to zero.
- Generally rates in terms of sample size n and a notion of $\text{size}(\mathcal{F})$.

Binary classification

- Take $\mathcal{Y} = \{0, 1\}$ and consider zero-one loss.
- Bayes rule can be written as

$$f^*(x) = \begin{cases} 1 & \eta(x) \geq 1/2 \\ 0 & \eta(x) < 1/2 \end{cases},$$

where $\eta(x) = \mathbb{E}[Y \mid X = x] = \mathbb{P}(Y = 1 \mid X = x)$ is the regression function.

- The choice for $m(x) = 1/2$ could be any of 0 or 1.
- Optimal Bayes risk can be written as

$$R^* = R(f^*) = \mathbb{E}|f^*(X) - \eta(X)| \quad (1)$$

i.e., the $L_1(\mathbb{P}_X)$ distance between η and its thresholded version.

- Another expression:

$$R^* = R(f^*) = \mathbb{E} \min\{\eta(X), 1 - \eta(X)\}$$

i.e. $R^* = \int \min\{\eta, 1 - \eta\} dP_X$. (Exercise)

- To see (1) note that

$$\begin{aligned}\mathbb{P}(Y \neq f^*(X) \mid X) &= \begin{cases} \mathbb{P}(Y = 0 \mid X) & f^*(X) = 1 \\ \mathbb{P}(Y = 1 \mid X) & f^*(X) = 0 \end{cases} \\ &= \begin{cases} f^*(X) - \eta(X) & f^*(X) = 1 \\ \eta(X) - f^*(X) & f^*(X) = 0 \end{cases} = |f^*(X) - \eta(X)|.\end{aligned}$$

- Let us compute the excess risk.

Theorem 1

The excess risk for binary classification is:

$$\begin{aligned}\mathcal{E}(f) &= \mathbb{E}[|2\eta(X) - 1||f^*(X) - f(X)|] \\ &= \mathbb{E}[|2\eta(X) - 1|1\{f^*(X) \neq f(X)\}]\end{aligned}$$

that is, $\mathcal{E}(f) = \int |f^* - f||2\eta - 1|d\mathbb{P}_X = \int |f^* - f|d\mathbb{Q} = \|f^* - f\|_{L_1(\mathbb{Q})}$.

- Here, $\mathbb{Q}$ is measure that has density $|2\eta - 1|$ w.r.t. $\mathbb{P}_X$.

- For two binary variables Y, Z , we have the following identities

$$\begin{aligned} 1\{Y \neq Z\} &= |Y - Z| = (Y - Z)^2 = Y(1 - Z) + Z(1 - Y) \\ &= Y + Z - 2YZ \end{aligned}$$

- Thus, for any $f : \mathcal{X} \rightarrow \mathcal{Y} = \{0, 1\}$, we have

$$1\{Y \neq f(X)\} = Y + f(X) - 2Yf(X)$$

- Subtracting similar identity for $1\{Y \neq f^*(X)\}$, we obtain

$$1\{Y \neq f(X)\} - 1\{Y \neq f^*(X)\} = (2Y - 1)(f^*(X) - f(X)) \quad (2)$$

- Taking expectation of both sides:

$$\begin{aligned} R(f) - R(f^*) &= \mathbb{E}[(2Y - 1)(f^*(X) - f(X))] \\ &\stackrel{(a)}{=} \mathbb{E}[(2\eta(X) - 1)(f^*(X) - f(X))] \\ &\stackrel{(b)}{=} \mathbb{E}[|2\eta(X) - 1||f^*(X) - f(X)|] \end{aligned}$$

Exercise: Justify steps (a) and (b).

Yet another expression

- From (2), we have

$$|1\{Y \neq f(X)\} - 1\{Y \neq f^*(X)\}| = |f^*(X) - f(X)| \quad (3)$$

- Hence, with $\ell_f(Z) := 1\{Y \neq f(X)\}$ where $Z = (X, Y)$, we have

$$\mathcal{E}(f) = \mathbb{E}[|2\eta(X) - 1| |\ell_f(Z) - \ell_{f^*}(Z)|] \quad (4)$$

Plugin classifier

- Form of $f^*(x) = 1\{\eta(x) \geq 1/2\}$ suggest the plugin estimator

$$f_{\hat{\eta}}(x) = 1\{\hat{\eta}(x) \geq 1/2\}$$

where $\hat{\eta}$ is an estimator of η .

Theorem 2

For any $\hat{\eta} : \mathcal{X} \rightarrow \mathbb{R}$, we have $\mathcal{E}(f_{\hat{\eta}}) \leq 2\mathbb{E}|\eta(X) - \hat{\eta}(X)|$.

i.e., $\mathcal{E}(f_{\hat{\eta}}) \leq 2\|\eta - \hat{\eta}\|_{L^1(\mathbb{P}_X)}$.

Proof:

- Use Theorem 1.
- $f^* \neq f_{\hat{\eta}}$ iff η and $\hat{\eta}$ are on different sides of $1/2$.
- In that case, $|\eta - 1/2| \leq |\eta - \hat{\eta}|$.
- Thus, $1\{f^* \neq f_{\hat{\eta}}\}|\eta - 1/2| \leq |\eta - \hat{\eta}|$. Integrate w.r.t. $\mathbb{P}_X$.



- Having a good regression estimator $\hat{\eta}$ give us a good plugin classifier.
- It is enough for $\hat{\eta}$ to be consistent η in $L^1(\mathbb{P}_X)$ norm (or $L^2(\mathbb{P}_X)$):

$$\mathcal{E}(f_{\hat{\eta}}) \leq 2\|\eta - \hat{\eta}\|_{L^1(\mathbb{P}_X)} \leq 2\|\eta - \hat{\eta}\|_{L^2(\mathbb{P}_X)}$$

(In a probability space, L^2 norm dominates L^1 : $\mathbb{E}|Z| \leq \sqrt{\mathbb{E}Z^2}$)

- However, having consistent estimate of η is not necessary.
- Excess risk can be much smaller than $2\|\eta - \hat{\eta}\|_{L^1(\mathbb{P}_X)}$.

Example 3

- Assume $\eta \in \{0, 1\}$.
- Then, there is always $\hat{\eta}$ satisfying
 - η and $\hat{\eta}$ are on the same side of $1/2$.
 - $|\hat{\eta}(X) - \eta(X)| = (1 - \varepsilon)/2$ a.s.
- We have $\mathcal{E}(f_{\hat{\eta}}) = 0$ while $2\|\eta - \hat{\eta}\|_{L^1(\mathbb{P}_X)} = 1 - \varepsilon$.

Plugin (parametric approach)

- We can directly model $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$ using a parametric model.
- Modeling $\eta(x)$ directly is sometimes called discriminative (as opposed to generative) approach.
- For example, in **logistic regression**, we assume

$$\mathbb{P}(Y = 1 \mid X = x) = \sigma(\theta^T x), \quad \sigma(t) = \frac{1}{1 + e^{-t}}$$

- Can estimate θ based on a training sample, using e.g. MLE $\hat{\theta}$.
- Then, $\hat{\eta}(x) = \sigma(\hat{\theta}^T x)$ and the plugin rule will be

$$f_{\hat{\eta}}(x) = 1\{\hat{\eta}(x) \geq 1/2\} = 1\{\hat{\theta}^T x \geq 0\}. \quad (5)$$

or $x \mapsto \text{sign}(\hat{\theta}^T x)$ if we work with $\mathcal{Y} = \{-1, 1\}$.

- Decision boundary in (5) is linear.

- With logistic loss $\ell_{\text{lg}}(y, t) = -yt + \log(1 + e^t)$, the MLE is

$$\hat{\theta} \in \operatorname{argmin}_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \ell_{\text{lg}}(Y_i, \theta^T x_i)$$

- Equivalently, with $\mathcal{F}_{\text{lin}} := \{x \mapsto \theta^T x : \theta \in \mathbb{R}^d\}$

$$\hat{\eta} \in \operatorname{argmin}_{f \in \mathcal{F}_{\text{lin}}} \frac{1}{n} \sum_{i=1}^n \ell_{\text{lg}}(Y_i, f(X_i))$$

- I.e., logistic regression is ERM with a surrogate loss function (replacing 0-1 with the smooth logistic loss).
- Logistic loss $t \mapsto \ell_{\text{lg}}(y, t)$ is convex.
- Can obtain the solution by Newton's method (leads to iteratively reweighed least-squares).
- Can perform SGD which looks like a variant of the perceptron algorithm.

Plugin (generative approach)

- Start with class-conditional densities $p_k(x) = p(x \mid Y = k)$,
- and class priors $\pi_k = \mathbb{P}(Y = k)$, so that

$$\mathbb{P}(Y = k \mid X = x) = \frac{\pi_k p_k(x)}{\sum_{\ell=1}^K \pi_\ell p_\ell(x)}.$$

- Can estimate p_k and π_k from training data.
- E.g., fit a kernel density estimate to all $\{X_i : Y_i = k\}$ to get $\hat{p}_k$.
- $\hat{\pi}_k = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{Y_i = k\}$.
- Optimal Bayes classification rule can be written as

$$f^*(x) = \operatorname{argmax}_{k \in [K]} \pi_k p_k(x).$$

- Any collection of function $\delta_k(x)$, $k = 1, \dots, K$ for which $f^*(x) = \operatorname{argmax}_{k \in [K]} \delta_k(x)$ are called discriminant functions.

- We could also postulate a parametric form for p_k , e.g.

$$X \mid Y = k \sim N(\mu_k, \Sigma_k) \quad (6)$$

and estimate μ_k and Σ_k from training data.

- Bayes rule in model (6) gives **quadratic discriminant analysis (QDA)**:

$$f^*(x) = \operatorname{argmax}_k \left\{ -\frac{1}{2}(x - \mu_k)^T \Sigma_k (x - \mu_k) + c_k \right\}, \quad (7)$$

where $c_k = -\frac{1}{2} \log |\Sigma_k| + \log \pi_k$. (Chapter 4 of ELS or HW1)

- When $\Sigma_k = \Sigma$ for all k , (7) simplifies to **linear discriminant analysis (LDA)**:

$$f^*(x) = \operatorname{argmax}_k (w_k^T x + b_k),$$

where $w_k = \Sigma^{-1} \mu_k$ and $b_k = -\frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$.

- Decision boundaries are linear in LDA.
- In binary classification, f^* of LDA is a thresholded linear function $f^*(x) = 1\{w^T x + b \geq 0\}$. or $f^*(x) = \operatorname{sign}(w^T x + b)$.

Fisher LDA

- A method of dimension reduction geared towards classification.
- Project X onto one dimension: $X_w := w^T X$.
- Classify the resulting scalar variable X_w .
- Choose w to maximize the ratio of *between* over *within* cluster variation:

$$w^* = \operatorname{argmax}_w \frac{\operatorname{var}(\mathbb{E}[X_w | Y])}{\mathbb{E} \operatorname{var}(X_w | Y)}$$

- This leads to a generalized eigenvalue problem (see HW1).
- In binary classification, there is a closed form: $w^* \propto \bar{\Sigma}^{-1}(\mu_1 - \mu_2)$.
- Equivalent formulation:

$$w^* = \operatorname{argmax}_w \frac{\operatorname{var}(\mathbb{E}[X_w | Y])}{\operatorname{var}(X_w)}$$

- Contrast with PCA which maximizes $w \mapsto \operatorname{var}(X_w)$. (No label info Y .)

Stochastic gradient descent (SGD)

- A general stochastic optimization technique for minimizing $\theta \mapsto \mathbb{E}L(\theta; Z)$.
- Assume that we have iid samples $Z_1, Z_2, \dots \sim Z$.
- SGD has the following update

$$\theta^{t+1} = \theta^t - \alpha \nabla L(\theta^t; Z_t)$$

- Usual (batch) gradient descent on the empirical loss $\theta \mapsto \frac{1}{n} \sum_{i=1}^n L(\theta; Z_i)$, has the following update

$$\theta^{t+1} = \theta^t - \alpha \frac{1}{n} \sum_{i=1}^n \nabla L(\theta^t; Z_t)$$

- In both cases the step we take is unbiased estimate of the gradient of $\mathbb{E}L(\theta; Z)$, the for SGD the variance is a lot higher, while the computation is a lot cheaper.

SGD for logistic regression and perceptron

- The SGD update for logistic regression ($Y_t \in \{0, 1\}$)

$$\theta_{t+1} = \theta_t + \alpha(Y_t - \sigma(\theta_t^T X_t))X_t$$

- Replacing $\sigma(\cdot)$ with hard threshold $1\{\cdot \geq 0\}$, gives the **perceptron algorithm**:

$$\theta^{t+1} = \theta^t + \alpha(Y_t - 1\{\theta_t^T X_t \geq 0\})X_t$$

- Alternative form with $Y_t \in \{-1, 1\}$ (and $\alpha = 1$):
- $\theta_0 = 0$, $t = 0$, and repeat while there are misclassified pairs:
 - Pick i (if any) such that $Y_i \neq \text{sign}(\theta_t^T X_i)$.
 - **Update:** $\theta_{t+1} \leftarrow \theta_t + Y_i X_i$.
 - Increment t .

Perceptron

- **Perceptron Algorithm:**

$\theta_0 = 0$, $t = 0$, and repeat while there are misclassified pairs:

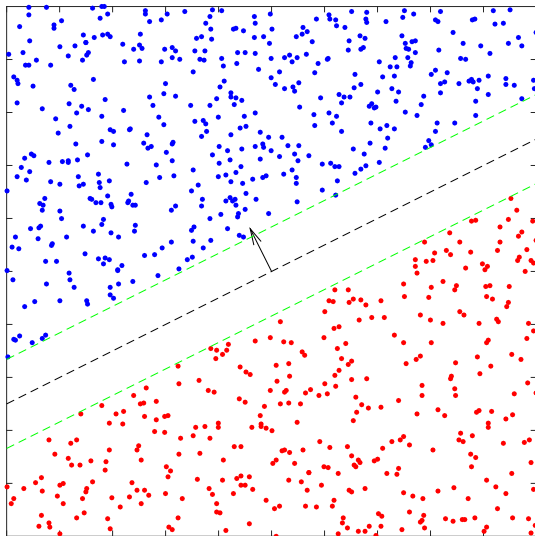
- Pick i (if any) such that $Y_i \neq \text{sign}(\theta_t^T X_i)$.
 - **Update:** $\theta_{t+1} \leftarrow \theta_t + Y_i X_i$.
 - Increment t .
- Data $(x_i, y_i) \in \mathbb{R}^d \times \{-1, 1\}$, $i = 1, \dots, n$ is **linearly separable** if there is a separating hyperplane:

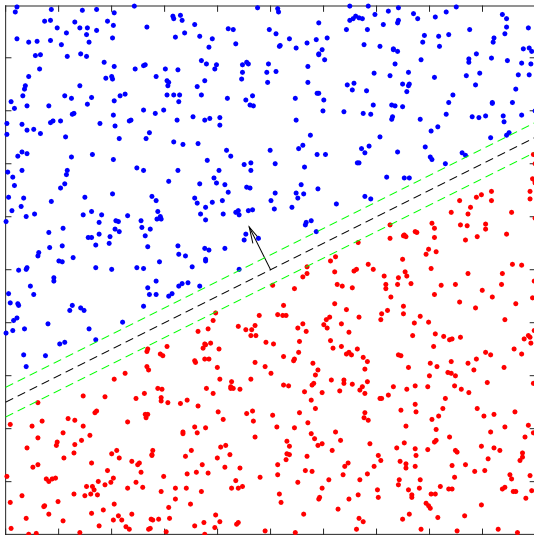
$$\exists \theta \in \mathbb{R}^d, \quad y_i(\theta^T x_i) > 0$$

in which case, we can define the margin:

$$\gamma = \min_i y_i \frac{\theta^T x_i}{\|\theta\|} \quad (8)$$

- θ need not be unique (even its direction).





Perceptron

- Assuming $\|\theta\| = 1$, the margin is $\gamma = \min_i y_i \theta^T x_i$.

Theorem 3

For any linearly separable data $\{(x_i, y_i)\} \subset \mathbb{R}^d \times \{-1, 1\}$ with margin γ as in (8) and radius $R = \max_i \|x_i\|$, the perceptron algorithm terminates in

at most R^2/γ^2 steps

and returns a separating hyperplane (for any choices made in the updates).

Assuming linear separation:

- Perceptron achieves zero empirical risk for the 0-1 loss.
- Exact algorithm for solving ERM with 0-1 loss over **thresholded linear function** class

$$\min_{f \in \mathcal{F}_{\text{t-lin}}} \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{Y_i \neq f(X_i)\}, \quad \mathcal{F}_{\text{t-lin}} := \{x \mapsto \text{sign}(\theta^T x) : \theta \in \mathbb{R}^d\}$$

Proof

- WLOG, assume that “true” θ is normalized: $\|\theta\| = 1$.
- Idea of the proof: $\langle \theta, \theta_t \rangle \gtrsim t$ while $\|\theta_t\| \lesssim \sqrt{t}$.
- Recalling $\theta_{t+1} = \theta_t + y_i x_i$, and the definition of margin:

$$\langle \theta_{t+1}, \theta \rangle - \langle \theta_t, \theta \rangle = y_i \langle x_i, \theta \rangle \geq \gamma$$

- Hence, $\langle \theta_t, \theta \rangle \geq \gamma t$ since $\theta_0 = 0$.
- On the other hand

$$\begin{aligned}\|\theta_{t+1}\|^2 &= \|\theta_t\|^2 + \|x_i\|^2 + 2y_i \langle \theta_t, x_i \rangle \\ &\leq \|\theta_t\|^2 + R^2.\end{aligned}$$

- It follows that $\|\theta_t\|^2 \leq tR^2$.
- Cauchy-Schwarz gives:

$$\gamma t \leq \langle \theta, \theta_t \rangle \leq \|\theta_t\| \leq \sqrt{t}R. \quad \square$$

- Exercise: Show that if we start with $\theta_0 \neq 0$, we have the bound $\sqrt{t} \leq (R + \sqrt{R^2 + 4b\gamma})/(2\gamma) \leq R/\gamma + \sqrt{b/\gamma}$ where $b = \|\theta_0\| - \langle \theta_0, \theta \rangle$.

Perception, general inner product

- Can write θ_t as (weighted) combination of data points:

$$\theta_t = \sum_{j=1}^n \alpha_j^{(t)} x_j, \quad \|\alpha^{(t)}\|_1 = \sum_i |\alpha_j^{(t)}| = t$$

- Originally all the weights are zero; perception gradually updates them.
- **Perceptron Algorithm:**

$\alpha^{(0)} = 0$, $t = 0$, and repeat while there are misclassified pairs:

- Pick i (if any) such that $y_i \neq \hat{y}_i^{(t)}$ where

$$\hat{y}_i^{(t)} := \text{sign} \left(\sum_{j=1}^n \alpha_j^{(t)} \langle x_j, x_i \rangle \right)$$

- **Update:** $\alpha_i^{(t+1)} := \alpha_i^{(t)} + y_i$. Increment t .
- Perceptron (and Theorem 3) works for general *inner product* $\langle x_i, x_j \rangle$.
- Can even replace inner product with kernels $K(x_i, x_j)$. [Linear separation in feature space.]

A bit of notational overhead

- (Excess) risk also depends on P the unknown distribution of (X, Y) :

$$\begin{aligned}\mathcal{E}(f; P) &= R(f; P) - \inf_f R(f; P) \\ &= R(f; P) - R^*(P)\end{aligned}$$

which itself can be thought of as a loss between f and P .

A bit of notational overhead

- Let $Z^n = (Z_i, i = 1, \dots, n) \in \mathcal{Z}^n$ be the training data where

$$Z_i = (X_i, Y_i) \in \mathcal{Z} := \mathcal{X} \times \{-1, 1\} \subset \mathbb{R}^d \times \{-1, 1\}.$$

- A classification rule or algorithm is specified by a collection of function $\hat{f}_n : \mathcal{X} \times \mathcal{Z}^n \rightarrow \{-1, 1\}$ for $n = 1, 2, \dots$.
- A trained classifier is a random function $x \mapsto \hat{f}_n(x; Z^n)$, with randomness due to training data Z^n .
- We also write $\hat{f}_n = \hat{f}_n(\cdot; Z^n)$ or $\hat{f}_n(x) = \hat{f}_n(x; Z^n)$ for simplicity.
- Notational overload: $\hat{f}_n : \mathcal{X} \rightarrow \{-1, 1\}$ is a random function or $\hat{f}_n : \mathcal{X} \times \mathcal{Z}^n \rightarrow \{-1, 1\}$ is a deterministic function depending on the context.

A bit of notational overhead

- The excess risk of $\hat{f}_n$ is the random quantity $\mathcal{E}(\hat{f}_n; P)$.
- We can look at $\mathbb{E}[\mathcal{E}(\hat{f}_n; P)]$ which is a form of risk itself (in the decision theoretic sense). Here expectation is w.r.t. randomness in Z^n .
- Let us call $\mathbb{E}[\mathcal{E}(\hat{f}_n; P)]$ the **meta excess risk**. (Non-standard terminology.)
- Meta risk measures the quality of the classification rule $\hat{f}_n : \mathcal{X} \times \mathcal{Z}^n \rightarrow \{-1, 1\}$ relative to the probability distribution P on $\mathcal{Z}$.

A bit of notational overhead

- We can also write $\mathbb{E}_{Z^n \sim P^n}[\mathcal{E}(\hat{f}_n; P)]$ or just $\mathbb{E}_{P^n}[\mathcal{E}(\hat{f}_n; P)]$ to emphasize.
- Consider binary classification with 0-1 loss:

$$R(f; P) = \mathbb{E}_P 1\{f(X) \neq Y\}$$

where $\mathbb{E}_P$ or $\mathbb{E}_{Z \sim P}$ is to emphasize that the expectation is w.r.t $Z \sim P$.

- The meta risk is

$$\begin{aligned}\mathbb{E}[R(\hat{f}_n; P)] &= \mathbb{E}_{P^n} \mathbb{E}_P [1\{\hat{f}_n(X; Z^n) \neq Y\}] \\ &= \mathbb{E}_{P^{n+1}} [1\{\hat{f}_n(X; Z^n) \neq Y\}] \\ &= \mathbb{P}_{P^{n+1}}(\hat{f}_n(X) \neq Y)\end{aligned}$$

where $\mathbb{E}_{P^{n+1}}$ or $\mathbb{E}_{(Z, Z^n) \sim P^{n+1}}$ is to emphasize underlying distribution.

- We can simply just write $\mathbb{E}[R(\hat{f}_n; P)] = \mathbb{P}(\hat{f}_n(X) \neq Y)$.
- Meta risk is the probability of misclassification taking into account all the sources of randomness. (Can think of it as the limit of “ideal” leave-one out cross validation on N independent batches of size $n + 1$ as $N \rightarrow \infty$.)

Perceptron excess risk

- Consider a randomized version (somewhat like a regularization):
- Let $\hat{f}_m$ be the classifier returned by perceptron when trained on $Z^m = (Z_1, \dots, Z_m)$ for $m \leq n$.
- Let $\tilde{f}_n$ be chosen at random from $\{\hat{f}_1, \hat{f}_2, \dots, \hat{f}_n\}$.

Theorem 4

Assume that P is such that $\exists \theta \in S^{d-1}$ and $\gamma > 0$, such that

$$\|X\| \leq R, \quad \text{and, } (\theta^T X)Y \geq \gamma \quad \text{a.s.}$$

Then, the excess risk of $\tilde{f}_n$ (for 0-1 loss) is

$$\mathbb{E}[\mathcal{E}(\tilde{f}_n, P)] = \mathbb{E}[R(\tilde{f}_n, P)] \leq \frac{R^2}{n\gamma^2}$$

(The equality is since $R^*(P) = 0$ in this case.)

Proof

- Write $R(f) = R(f, P)$ for simplicity.

$$\begin{aligned}\mathbb{E}R(\tilde{f}_n) &= \frac{1}{n} \sum_{m=1}^n \mathbb{E}R(\hat{f}_m) \\ &= \frac{1}{n} \sum_{m=1}^n \mathbb{P}(\hat{f}_m(X) \neq Y) = \frac{1}{n} \sum_{m=1}^n \mathbb{P}(\hat{f}_m(X_{m+1}) \neq Y_{m+1})\end{aligned}$$

- On the other hand,

$$\sum_{m=1}^n 1_{\{\hat{f}_m(X_{m+1}) \neq Y_{m+1}\}} \leq \frac{R^2}{\gamma^2}$$

since the LHS is the number of misclassifications in a single run of the perceptron algorithm on data Z^{n+1} , with the convention that we choose the smallest index among misclassified examples at each update.

A minimax lower bound

- Assume the 0-1 loss and consider the following class of probability distribution on $\mathcal{Z}$:

$$\mathcal{P} := \mathcal{P}(\mathcal{Z}) := \left\{ P : \inf_{f \in \mathcal{F}_{\text{t-lin}}} R(f, P) = 0 \right\}$$

- and the class of all *classification rules* based on a sample of size n :

$$\mathcal{C}_n := \mathcal{C}_n(\mathcal{X}) := \{ \hat{f}_n \mid \hat{f}_n : \mathcal{X} \times \mathcal{Z}^n \mapsto \{-1, 1\} \}$$

(Recall $\mathcal{Z} = \mathcal{X} \times \{-1, 1\}$)

Theorem 5

Assume $\mathcal{X} \subset \mathbb{R}^d$ has non-empty interior. Then,

$$\inf_{\hat{f}_n \in \mathcal{C}_n} \sup_{P \in \mathcal{P}} \mathbb{E}[\mathcal{E}(\hat{f}_n, P)] \geq \frac{\min(n, d) - 1}{2n} \left(1 - \frac{1}{n}\right)^n$$

- For any classification rule $\hat{f}_n : \mathcal{X} \times \mathcal{Z}^n \mapsto \{-1, 1\}$, there exists a probability distribution $P \in \mathcal{P}(\mathcal{Z})$ such that $\mathbb{E}[\mathcal{E}(\hat{f}_n, P)] \geq \dots$.

Proof

- We construct a random probability distribution $\mathbf{P}$ on $\mathcal{Z}$ and show that

$$\inf_{\hat{f}_n \in \mathcal{C}_n} \mathbb{E}[\mathcal{E}(\hat{f}_n, \mathbf{P})] \geq \frac{\min(n, d) - 1}{2n} \left(1 - \frac{1}{n}\right)^n$$

where $\mathbb{E}$ is also over the randomness of $\mathbf{P}$.

- The result follows since maximum risk is lower bounded by the average (Bayes) risk. (Also called the *probabilistic approach*.)
- In fact, we pick $\mathbf{P}$ uniformly at random from $\mathcal{P}_0 \subset \mathcal{P}$, so we have the stronger result:

$$\inf_{\hat{f}_n \in \mathcal{C}_n} \sup_{P \in \mathcal{P}_0} \mathbb{E}[\mathcal{E}(\hat{f}_n, P)] \geq \inf_{\hat{f}_n \in \mathcal{C}_n} \mathbb{E}[\mathcal{E}(\hat{f}_n, \mathbf{P})]$$

Proof (constructing P)

- Pick linearly independent $\{v_1, \dots, v_d\} \subset \mathcal{X}$ and define:

$$P_b = \sum_{j=1}^d w_j \delta_{(v_j, b_j)}, \quad w_j = \begin{cases} \frac{\varepsilon}{d-1} & j = 1, \dots, d-1 \\ 1 - \varepsilon & j = d \end{cases}$$

where $b = (b_1, \dots, b_d) \in \{-1, 1\}^d$.

- For any P_b the Bayes risk over t-lin. class is zero: there exists a linear function f_b such that $f_b(v_j) = b_j$ for all $j = 1, \dots, d$ for any b .
- The key is that in a draw from P_b we mostly see (v_d, b_d) .
- Let $B = (B_1, \dots, B_d) \sim \text{Unif}(\{-1, 1\}^d)$ and define $\mathbf{P} = P_B$.
- That is, we draw the data as

$$B \sim \text{Unif}(\{-1, 1\}^d)$$

$$(X, Y), (X_1, Y_1), \dots, (X_n, Y_n) \mid B = b \stackrel{\text{iid}}{\sim} P_b,$$

- In other words, $Z, Z_1, \dots, Z_n \mid B = b \sim P_b$.

- Let $V(X^n) = \{v_1, \dots, v_{d-1}\} \setminus \{X_1, \dots, X_n\}$, i.e., unseen “light” examples.
- Let $J = J(X^n)$ be indices of elements of $V(X^n)$. The key is

$$B_J \mid |J| = k \sim \text{Unif}(\{-1, 1\}^k)$$

A bit more precisely:

$$\begin{aligned} B_J \mid Z^n = z^n, |J| = k &\sim \text{Unif}(\{-1, 1\}^k) \\ B_{J^c} \mid Z^n = z^n &\sim \text{deterministic} \end{aligned}$$

(Exercise: Argue this and fill in the details.)

- Recall that $\mathbb{E}\mathcal{E}(\hat{f}_n, \mathbf{P}) = \mathbb{E}R(\hat{f}_n, \mathbf{P}) = \mathbb{P}(\hat{f}_n(X) \neq Y)$,

$$\begin{aligned}\mathbb{P}(\hat{f}_n(X) \neq Y \mid B, Z^n = z^n) &= \sum_j w_j 1\{\hat{f}_n(v_j, z^n) \neq B_j\} \\ &\geq \sum_{j \in J(x^n)} w_j 1\{\hat{f}_n(v_j, z^n) \neq B_j\} \\ &= \frac{\varepsilon}{d-1} \sum_{j \in J(x^n)} 1\{\hat{f}_n(v_j, z^n) \neq B_j\}\end{aligned}$$

- Taking the expectation w.r.t. conditional distribution of B given $Z^n = z^n$:

$$\begin{aligned}\mathbb{P}(\hat{f}_n(X) \neq Y \mid Z^n = z^n) &= \frac{\varepsilon}{d-1} \sum_{j \in J(x^n)} \mathbb{E}[1\{\hat{f}_n(v_j, z^n) \neq B_j\} \mid Z^n = z^n] \\ &= \frac{\varepsilon}{d-1} \sum_{j \in J(x^n)} \mathbb{P}(\hat{f}_n(v_j, z^n) \neq B_j \mid Z^n = z^n) \\ &= \frac{\varepsilon}{d-1} \sum_{j \in J(x^n)} \frac{1}{2} \\ &= \frac{1}{2} \frac{\varepsilon}{d-1} |J(x^n)|.\end{aligned}$$

- In other words, the performance of the decision rule on B_J (the labels of unseen examples from the light pairs) is equivalent to random guessing.
- Thus, we have shown

$$\mathbb{P}(\hat{f}_n(X) \neq Y \mid Z^n) \geq \frac{\varepsilon}{2(d-1)} |J(X^n)|. \quad (9)$$

- Using smoothing (since $|J|$ is a function of Z^n) it follows that

$$\mathbb{P}(\hat{f}_n(X) \neq Y \mid |J|) \geq \frac{\varepsilon}{2(d-1)} |J|.$$

- Either from this or from (9) directly, by smoothing

$$\mathbb{P}(\hat{f}_n(X) \neq Y) \geq \frac{\varepsilon}{2(d-1)} \mathbb{E}|J|$$

- But we have

$$\mathbb{E}|J| = \sum_{j=1}^{d-1} \mathbb{P}(v_j \notin \{X_1, \dots, X_n\}) = (d-1) \left(1 - \frac{\varepsilon}{d-1}\right)^n$$

- Putting the pieces together

$$\mathbb{E}\mathcal{E}(\hat{f}_n, \mathbf{P}) \geq \frac{\varepsilon}{2} \left(1 - \frac{\varepsilon}{d-1}\right)^n$$

- Choose $\varepsilon = \frac{\min(n, d)-1}{n}$ and note that

$$\frac{\min(n, d) - 1}{n(d-1)} \leq \frac{1}{n}.$$

Fast rates under (strong) margin condition

- Recall that $\eta(X) = \mathbb{P}(Y = 1 \mid X = x)$ and f^* is the optimal Bayes classifier.

Theorem 6

Assume that $|2\eta(X) - 1| \geq \gamma$ almost surely, for some $\gamma > 0$. Let $\hat{f}_n$ be the ERM with zero-one loss over a finite function class $\mathcal{F}$, and $f^ \in \mathcal{F}$.*

Then, with probability at least $1 - \delta$,

$$\mathcal{E}(\hat{f}_n) \leq C_\gamma \frac{\log(|\mathcal{F}|/\delta)}{n},$$

where $C_\gamma > 0$ is a constant only dependent on γ .

- Proof uses concentration inequalities.

- An example of a concentration inequality

Theorem 7 (Bernstein inequality)

Let $X_i, i = 1, \dots, n$ be independent, centered, bounded variables $|X_i| \leq b$ (a.s.), and let $S_n = \sum_{i=1}^n X_i$ and $v = \sum_{i=1}^n \mathbb{E}X_i^2$. Then,

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2v + 2bt/3}\right), \quad \forall t \geq 0.$$

- Note that $v = \text{var}(S_n)$.
- Same bound holds for $\mathbb{P}(S_n \leq -t)$.
- Thus $\mathbb{P}(|S_n| \geq t) \leq$ twice the bound.
- If $\mathbb{E}[X_i] = \mu_i$ while $|X_i - \mu_i| \leq b$ a.s., we apply the result to $X_i - \mu_i$.
- Letting $\mu = \mathbb{E}[S_n] = \sum_i \mu_i$, we get

$$\mathbb{P}(S_n - \mu \geq t) \leq \exp\left(-\frac{t^2}{2v + 2bt/3}\right), \quad \forall t \geq 0.$$

where $v = \text{var}(S_n) = \sum_i \mathbb{E}(X_i - \mu_i)^2$.

- An example of a concentration inequality

Theorem 8 (Bernstein inequality)

Let $X_i, i = 1, \dots, n$ be independent, centered, bounded variables $|X_i| \leq b$ (a.s.), and let $S_n = \sum_{i=1}^n X_i$ and $v = \sum_{i=1}^n \mathbb{E}X_i^2$. Then,

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{t^2}{2v + 2bt/3}\right), \quad \forall t \geq 0.$$

- Note that $v = \text{var}(S_n)$.
- Same bound holds for $\mathbb{P}(S_n \leq -t)$.
- Thus $\mathbb{P}(|S_n| \geq t) \leq \text{twice the bound}$.
- If $\mathbb{E}[X_i] = \mu_i$ while $|X_i - \mu_i| \leq b$ a.s., we apply the result to $X_i - \mu_i$.
- Letting $\bar{\mu} = \mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_i \mu_i$, we get

$$\mathbb{P}(\bar{X}_n - \mu \geq t) \leq \exp\left(-\frac{nt^2}{2\bar{v} + 2bt/3}\right), \quad \forall t \geq 0.$$

where $\bar{v} = \text{var}(\bar{X}_n) = \frac{1}{n} \sum_i \mathbb{E}(X_i - \mu_i)^2$.

Fast rates

- Taking $t = \beta\bar{v} + \varepsilon$, for $\beta, \varepsilon > 0$. Then, $\bar{v} \leq t/\beta$, and

$$\begin{aligned}\mathbb{P}(\bar{X}_n - \bar{\mu} \geq t) &\leq \exp\left(-\frac{nt}{2/\beta + 2b/3}\right) \\ &\leq \exp(-cnt) \\ &\leq \exp(-cn\varepsilon)\end{aligned}$$

- That is, with $c = (2/\beta + 2b/3)^{-1}$,

$$\mathbb{P}(\bar{X}_n \geq \bar{\mu} + \beta\bar{v} + \varepsilon) \leq \exp(-cn\varepsilon), \quad \forall \varepsilon > 0$$

and similarly,

$$\mathbb{P}(\bar{X}_n \leq \bar{\mu} - \beta\bar{v} - \varepsilon) \leq \exp(-cn\varepsilon), \quad \forall \varepsilon > 0$$

- Thus,

$$\bar{X}_n \in (\bar{\mu} - \beta\bar{v}, \bar{\mu} + \beta\bar{v}) + [-\varepsilon, \varepsilon]$$

for $\varepsilon = O\left(\frac{\log(1/\delta)}{n}\right)$ w.p. $\geq 1 - 2\delta$.

Fast rates (variance bounded by mean)

- Assume that $\bar{v} \leq C\bar{\mu}$ for $C > 0$. (Note that this requires $\bar{\mu} \geq 0$.)
- Let $\beta = \alpha/C$, for some $\alpha > 0$, so that $\beta\bar{v} \leq \alpha\bar{\mu}$.
- It follows that, with $c = (2C/\alpha + 2b/3)^{-1}$,

$$\mathbb{P}(\bar{X}_n \geq (1 + \alpha)\bar{\mu} + \varepsilon) \leq \exp(-cn\varepsilon), \quad \forall \varepsilon > 0, \quad (10)$$

$$\mathbb{P}(\bar{X}_n \leq (1 - \alpha)\bar{\mu} - \varepsilon) \leq \exp(-cn\varepsilon), \quad \forall \varepsilon > 0. \quad (11)$$

- Thus,

$$\bar{X}_n \in ((1 - \alpha)\bar{\mu}, (1 + \alpha)\bar{\mu}) + [-\varepsilon, \varepsilon]$$

for $\varepsilon = O(\frac{\log(1/\delta)}{n})$ w.p. $\geq 1 - 2\delta$.

- Contrast with the general bound

$$\bar{X}_n \in \bar{\mu} + [-\varepsilon, \varepsilon]$$

for $\varepsilon = O(\sqrt{\frac{\bar{v} \log(1/\delta)}{n}})$ w.p. $\geq 1 - 2\delta$, for large n .

Proof of Theorem 6

- Let $\ell_f(z) = 1\{f(x) \neq y\}$ for $z = (x, y)$.
- As f varies in some class $\mathcal{F}$, it carves out the loss class $\{\ell_f : f \in \mathcal{F}\}$.
- We have $\mathcal{E}(f) = \mathbb{E}[\ell_f(Z) - \ell_{f^*}(Z)]$, and

$$\widehat{\mathcal{E}}(f) := \widehat{R}_n(f) - \widehat{R}_n(f^*) = \frac{1}{n} \sum_{i=1}^n [\ell_f(Z_i) - \ell_{f^*}(Z_i)]$$

- Note that $\mathbb{E}\widehat{\mathcal{E}}(f) = \mathcal{E}(f)$.
- To simplify further, let $\Delta_f(z) := \ell_f(z) - \ell_{f^*}(z)$. Then,

$$\mathcal{E}(f) = \mathbb{E}[\Delta_f(Z)], \quad \widehat{\mathcal{E}}(f) = \frac{1}{n} \sum_{i=1}^n \Delta_f(Z_i)$$

- $\Delta_f(Z_i) \in \{-1, 0, 1\}$ but $\mathbb{E}\Delta_f(Z_i) = \mathcal{E}(f) \geq 0$.

- Recall from (4) that $\mathcal{E}(f) = \mathbb{E}[|\Delta_f(Z)| |2\eta(X) - 1|]$.
- Under margin condition $|2\eta(X) - 1| \geq \gamma$ a.s., we have

$$\mathcal{E}(f) \geq \gamma \mathbb{E}|\Delta_f(Z)| \geq \gamma \mathbb{E}[\Delta_f(Z)]^2 \geq \gamma \text{var}(\Delta_f(Z))$$

and note that $\mathcal{E}(f) = \mathbb{E}[\Delta_f(Z)]$.

- In our previous notation, $\bar{v} \leq \gamma^{-1} \bar{\mu}$. Apply (11)

$$\mathbb{P}(\hat{\mathcal{E}}(f) \leq (1 - \alpha)\mathcal{E}(f) - \varepsilon) \leq \exp(-cn\varepsilon), \quad \varepsilon > 0$$

where $c = (\frac{2}{\alpha\gamma} + \frac{2}{3})^{-1}$.

- Union bound

$$\mathbb{P}(\exists f \in \mathcal{F} : \hat{\mathcal{E}}(f) \leq (1 - \alpha)\mathcal{E}(f) - \varepsilon) \leq |\mathcal{F}| \exp(-cn\varepsilon), \quad \varepsilon > 0$$

- Taking $\varepsilon = \frac{\log(|\mathcal{F}|/\delta)}{cn}$, w.p. $\geq 1 - \delta$,

$$\hat{\mathcal{E}}(f) \geq (1 - \alpha)\mathcal{E}(f) - \frac{\log(|\mathcal{F}|/\delta)}{cn}, \quad \forall f \in \mathcal{F}$$

- Rearranging, with $c' = (1 - \alpha)c$,

$$\mathcal{E}(f) \leq \frac{1}{1 - \alpha} \hat{\mathcal{E}}(f) + \frac{\log(|\mathcal{F}|/\delta)}{c'n}, \quad \forall f \in \mathcal{F}$$

w.p. $\geq 1 - \delta$.

- Applying to $\hat{f}_n$ and noting that $\hat{\mathcal{E}}(\hat{f}_n) \leq 0$ (why?),

$$\mathcal{E}(\hat{f}_n) \leq \frac{\log(|\mathcal{F}|/\delta)}{c'n}$$

w.p. $\geq 1 - \delta$. This is the desired result.

General (slow) rates

- Let $\mathcal{E}_{\mathcal{F}}(\cdot)$ be the excess risk over function class $\mathcal{F}$, i.e.,

$$\mathcal{E}_{\mathcal{F}}(f) = R(f) - \inf_{f \in \mathcal{F}} R(f).$$

Theorem 9

Let $\hat{f}_n$ be the ERM with zero-one loss over a finite function class $\mathcal{F}$.

Then, with probability at least $1 - 2\delta$,

$$\mathcal{E}_{\mathcal{F}}(\hat{f}_n) \leq C \sqrt{\frac{\log(|\mathcal{F}|/\delta)}{n}},$$

where C is a universal constant.

- Proof uses concentration inequalities.
- Theorem 9 holds for any “bounded” loss function (with the same proof).

- An example of a concentration inequality

Theorem 10 (Hoeffding inequality)

Let $X_i, i = 1, \dots, n$ be independent random variables with $X_i \in [a_i, b_i]$ a.s., and let $S_n = \sum_{i=1}^n X_i$. Then,

$$\mathbb{P}(S_n - \mathbb{E}S_n \geq t) \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right), \quad \forall t \geq 0.$$

- Same bound holds for the lower tail.
- If $\mathbb{E}[X_i] = \mu_i$ while $|X_i - \mu_i| \leq b$ a.s., we apply the result to $X_i - \mu_i$.
- Letting $\bar{\mu} = \mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_i \mu_i$, we get

$$\mathbb{P}(\bar{X}_n - \bar{\mu} \geq t) \leq \exp\left(-\frac{nt^2}{2b^2}\right), \quad \forall t \geq 0.$$

Proof of Theorem 9

- Let $\hat{f} = \hat{f}_n$ be the ERM over $\mathcal{F}$.
- Let $f_{\mathcal{F}}^* = \operatorname{argmin}_{f \in \mathcal{F}} R(f)$. We have

$$\begin{aligned}\mathcal{E}_{\mathcal{F}}(\hat{f}) &:= R(\hat{f}) - R(f_{\mathcal{F}}^*) \\ &\leq R(\hat{f}) - \hat{R}_n(\hat{f}) + \hat{R}_n(f_{\mathcal{F}}^*) - R(f_{\mathcal{F}}^*) \\ &\leq 2 \sup_{f \in \mathcal{F}} |\hat{R}_n(f) - R(f)| \\ &:= 2\|\hat{R}_n - R\|_{\mathcal{F}}\end{aligned}$$

- Recall that

$$\hat{R}_n(f) - R(f) = \frac{1}{n} \sum_{i=1}^n [\ell_f(Z_i) - \mathbb{E}\ell_f(Z_i)]$$

- For 0-1 loss: $\ell_f(Z_i) = 1\{f(X_i) \neq Y_i\}$

- We have $|\ell_f(Z_i) - \mathbb{E}\ell_f(Z_i)| \leq 1$, hence

$$\mathbb{P}\left(|\hat{R}_n(f) - R(f)| \geq t\right) \leq 2 \exp\left(-\frac{nt^2}{2}\right).$$

- Apply union bound

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |\hat{R}_n(f) - R(f)| \geq t\right) \leq 2|\mathcal{F}| \exp\left(-\frac{nt^2}{2}\right).$$

- Take $t = \sqrt{2 \log(|\mathcal{F}|/\delta)/n}$, w.p. $\geq 1 - 2\delta$

$$\|\hat{R}_n - R\|_{\mathcal{F}} \leq \sqrt{\frac{2 \log(|\mathcal{F}|/\delta)}{n}}.$$

- Hence, with the same probability

$$\mathcal{E}_{\mathcal{F}}(\hat{f}) \leq 2\sqrt{\frac{2 \log(|\mathcal{F}|/\delta)}{n}}.$$

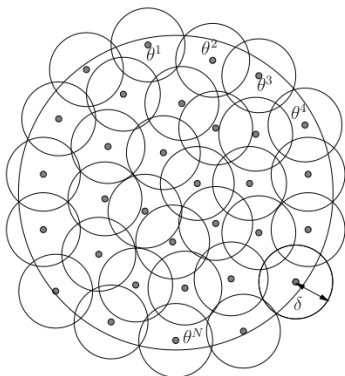
How to get around finiteness?

- Use discretization; leads to ε -net and metric entropy ideas.
- Use empirical process theory: reduce by symmetrization and concentration to bounding Rademacher averages. In this case, the size of the *trace* of $\ell_{\mathcal{F}}$ on $\{Z_1, \dots, Z_n\}$ matters, not the cardinality of $\mathcal{F}$ itself, i.e. the size of

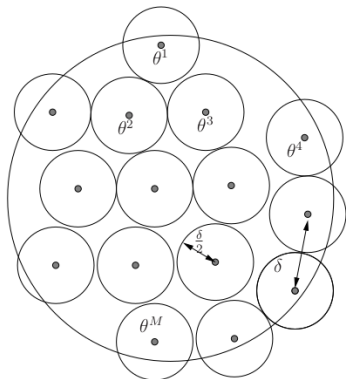
$$\ell_{\mathcal{F}}(Z^n) := \{(\ell_f(Z_1), \dots, \ell_f(Z_n)) : f \in \mathcal{F}\}.$$

Here $\ell_{\mathcal{F}} = \{\ell_f : f \in \mathcal{F}\}$ is the loss function class.

ε -net (or ε -cover) and metric entropy



(a)



(b)

(a) is a δ -covering and (b) a δ -packing².

²Picture from HDS book

General (slow) rate: via metric entropy

- Let $N = N(\varepsilon) = N(\varepsilon; \mathcal{F}, \|\cdot\|_\infty)$.
- Assume that $\mathcal{N} = \{f_1, \dots, f_N\}$ is an ε -cover of $\mathcal{F}$ in sup-norm:

$$\forall f \in \mathcal{F}, \quad \exists f_j \in \mathcal{N}, \quad \text{s.t.} \quad \|f - f_j\|_\infty \leq \varepsilon$$

- $\log N(\varepsilon)$ is called the **metric entropy**.

Theorem 11

Let $\hat{f}_n$ be the ERM with zero-one loss over a function class $\mathcal{F}$, and let $N(\varepsilon) = N(\varepsilon; \mathcal{F}, \|\cdot\|_\infty)$ be the ε -covering number of $\mathcal{F}$ in sup-norm.

Then, with probability at least $1 - 2\delta$,

$$\mathcal{E}_{\mathcal{F}}(\hat{f}_n) \lesssim \sqrt{\frac{\log(N(\varepsilon)/\delta)}{n}} + \varepsilon,$$

- $a \lesssim b$ means: there is universal constant C such that $a \leq Cb$.
- Choose ε to balance the two terms, i.e. solve

$$\log N(\varepsilon) \asymp n\varepsilon^2$$

Parametric example

- We have $N(\varepsilon; \mathbb{B}_2^d, \|\cdot\|_2) \leq (1 + 2/\varepsilon)^d$.
- Or $N(\varepsilon; \mathbb{B}_2^d, \|\cdot\|_2) \leq (3/\varepsilon)^d$ if $\varepsilon \leq 1$.
- Consider the parametric function class

$$\mathcal{F} = \{x \mapsto \sigma(\theta^T x) : \theta \in \mathbb{B}_2^d\}$$

where $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is some L -Lipschitz function.

- Assuming that $\mathcal{X} \subset r \mathbb{B}_2^d$, i.e. $x \in \mathcal{X} \implies \|x\|_2 \leq r$,

$$N(Lr\varepsilon; \mathcal{F}, \|\cdot\|_\infty) \leq N(\varepsilon; \mathbb{B}_2^d, \|\cdot\|_2).$$

- Assuming that $\varepsilon \leq Lr$, $\log N(\varepsilon; \mathcal{F}, \|\cdot\|_\infty) \leq d \log \left(\frac{3Lr}{\varepsilon} \right)$
- Follows that w.p. $\geq 1 - 2\delta$,

$$\mathcal{E}_{\mathcal{F}}(\hat{f}_n) \lesssim \sqrt{\frac{d \log(3Lr/\varepsilon) + \log(1/\delta)}{n}} + \varepsilon,$$

- Follows that w.p. $\geq 1 - 2\delta$,

$$\mathcal{E}_{\mathcal{F}}(\hat{f}_n) \lesssim \sqrt{\frac{d \log(3Lr/\varepsilon) + \log(1/\delta)}{n}} + \varepsilon,$$

- Take $\varepsilon = Lr(n/d)^{-1/2}$ and $\delta = n^{-d}$, to get w.p. $\geq 1 - 2n^{-d}$,

$$\mathcal{E}_{\mathcal{F}}(\hat{f}_n) \leq C_{L,r} \sqrt{\frac{d \log(n/d)}{n}},$$

assuming $d \leq n$.

- In general, log factor can be removed with a more careful argument (e.g., chaining).
- Note that we could have taken $\varepsilon \asymp (n/d)^{-\alpha}$ for any $\alpha \geq 1/2$.

Proof of Theorem 11

- $\ell_f(z) = 1\{y \neq f(x)\}$.
- Recall from (3) that for any $f, g \in \mathcal{F}$, $|\ell_f(z) - \ell_g(z)| = |f(x) - g(x)|$.
- Let $\bar{R}_n = \hat{R}_n(f) - R(f)$ be the centered empirical risk.

$$\begin{aligned} |\bar{R}_n(f) - \bar{R}_n(g)| &\leq \frac{1}{n} \sum_{i=1}^n \left\{ |\ell_f(Z_i) - \ell_g(Z_i)| + \mathbb{E} |\ell_f(Z_i) - \ell_g(Z_i)| \right\} \\ &= \frac{1}{n} \sum_{i=1}^n \left\{ |f(X_i) - g(X_i)| + \mathbb{E} |f(X_i) - g(X_i)| \right\} \\ &\leq 2\varepsilon \end{aligned}$$

assuming $\|f - g\|_\infty \leq \varepsilon$. Hence,

$$\sup_{f \in \mathcal{F}} |\bar{R}_n(f)| \leq \sup_{g \in \mathcal{N}} |\bar{R}_n(g)| + 2\varepsilon$$

or in short $\|\bar{R}_n\|_{\mathcal{F}} \leq \|\bar{R}_n\|_{\mathcal{N}} + 2\varepsilon$.

- Using our previous results: $\|\bar{R}_n\|_{\mathcal{N}} \leq \sqrt{\frac{2 \log(|\mathcal{N}|/\delta)}{n}}$ w.p. $\geq 1 - 2\delta$.
- Recalling that $|\mathcal{N}| = N(\varepsilon)$, we have w.p. $\geq 1 - 2\delta$,

$$\mathcal{E}_{\mathcal{F}}(\hat{f}_n) \leq 2\|\bar{R}_n\|_{\mathcal{F}} \leq 2\sqrt{\frac{2 \log(N(\varepsilon)/\delta)}{n}} + 4\varepsilon.$$

Empirical process approach

- Define Rademacher complexity of the function class $\mathcal{F}$ as

$$\mathcal{R}_n(\mathcal{F}) = \mathbb{E}_{X, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \right]. \quad (12)$$

- $\varepsilon_i \stackrel{\text{iid}}{\sim} \text{Unif}(\{-1, 1\})$ independent of $X = (X_1, \dots, X_n)$.

Theorem 12

For any b -uniformly bounded class of functions $\mathcal{F}$, we have

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \leq 2\mathcal{R}_n(\mathcal{F}) + \delta$$

with probability $1 - \exp(-n\delta^2/2b^2)$.

Proof sketch:

- Use bounded difference inequality to show that $\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}}$ is concentrated around $\mathbb{E}\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}}$.
- Use a symmetrization argument to upper bound $\mathbb{E}\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}}$ by the Rademacher complexity.

Bounded difference inequality

Theorem 13

Suppose that $f : \mathcal{X}^n \rightarrow \mathbb{R}$ satisfies the *bounded difference property* with constants $\{c_i\}$, and let $X = (X_1, \dots, X_n)$ have independent coordinates. Then,

$$\mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right)$$

- Bounded difference property:

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i,$$

for all choices of $x_1, \dots, x_n$ and x'_i .

- We apply these type of result to the **loss function class**

$$\ell_{\mathcal{F}} = \{\ell_f : f \in \mathcal{F}\}$$

where $\ell_f(z) = \ell(f(x), y)$.

- Recall that $\bar{R}_n(f) := \hat{R}_n(f) - R(f)$ is the centered empirical risk.

Corollary 1

For any b -uniformly bounded loss function call $\ell_{\mathcal{F}}$, we have

$$\|\bar{R}_n\|_{\mathcal{F}} \leq 2\mathcal{R}_n(\ell_{\mathcal{F}}) + \delta$$

with probability $1 - \exp(-n\delta^2/2b^2)$.

- It is thus enough to bound the Rademacher complexity of the loss class:

$$\mathcal{R}_n(\ell_{\mathcal{F}}) = \mathbb{E}_{Z, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \ell_f(Z_i) \right| \right]$$

- Let us write $\mathcal{R}_n(\mathcal{F} \mid X_1^n)$ for the empirical Rademacher complexity:

$$\mathcal{R}_n(\mathcal{F} \mid X_1^n) := \mathbb{E}_\varepsilon \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \right]$$

- Alternatively, let $\mathcal{F}(X_1^n) = \{(f(X_1), \dots, f(X_n)) : f \in \mathcal{F}\}$, then

$$\mathcal{R}_n(\mathcal{F} \mid X_1^n) = \mathbb{E}_\varepsilon \left[\sup_{a \in \mathcal{F}(X_1^n)} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i a_i \right| \right]$$

- From now on, let $\mathcal{F}$ be a class of binary functions.
- Then, $|\mathcal{F}(X_1^n)|$ is at most 2^n . (Reduction to finite case.)
- We say that $\mathcal{F}$ **shatters** X_1^n if $|\mathcal{F}(X_1^n)| = 2^n$.
- In interesting cases, $|\mathcal{F}(X_1^n)|$ is much smaller than 2^n .

Bounding $\mathcal{R}_n(\mathcal{F} \mid X)$

- Problem reduces to bounding

$$\mathcal{R}(A) := \mathbb{E} \sup_{a \in A} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i a_i \right|$$

Lemma 1

For any $A \subset \mathbb{R}^n$, with $D^2(A) = \sup_{a \in A} \frac{1}{n} \|a\|_2^2$,

$$\mathbb{E} \sup_{a \in A} \frac{1}{n} \sum_{i=1}^n \varepsilon_i a_i \leq \sqrt{\frac{2D^2(A) \log |A|}{n}}$$

$$\mathcal{R}(A) := \mathbb{E} \sup_{a \in A} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i a_i \right| \leq \sqrt{\frac{2D^2 \log(2|A|)}{n}}$$

- Second one follows from the first by applying it to $A \cup -A$.
- The first: maximum of sub-Gaussian variables.

Theorem 14

Assume $\{X_i\}_{i=1}^n$ are zero-mean RVs, sub-Gaussian with parameter σ . Then,

$$\mathbb{E}\left[\max_{i=1,\dots,n} X_i\right] \leq \sqrt{2\sigma^2 \log n}, \quad \forall n \geq 1$$

- Proof of 14: Jensen's inequality on $e^{\lambda Z}$ where $Z = \max_i X_i$.
- Proof of Lemma 1:
 - Rademacher variables ε_i are sub-Gaussian with $\sigma_i = 1$.
 - Hence, $\frac{1}{n} \sum_i \varepsilon_i a_i$ is sub-Gaussian with squared parameter

$$\sigma_a^2 = \frac{1}{n^2} \sum_i a_i^2 = \frac{\|a\|_2^2}{n^2} \leq \frac{D^2(A)}{n}.$$

- Let us define³

$$m_{\mathcal{F}}(n) := \max\{|\mathcal{F}(x_1^n)| : x_1, \dots, x_n \in \mathcal{X}\}$$

that is, maximum size of the trace of $\mathcal{F}$ on any n points from $\mathcal{X}$.

- From the result just proved

$$\mathcal{R}_n(\mathcal{F} \mid X_1^n) \leq \sqrt{\frac{2D_{\mathcal{F}}^2 \log m_{\mathcal{F}}(n)}{n}}.$$

where

$$D_{\mathcal{F}}^2 = D^2(\mathcal{F}(X_1^n)) = \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f^2(X_i) \quad (13)$$

³Another notation: $|\mathcal{F}(x_1^n)|$ is also written as $\Delta^{\mathcal{F}}(A)$ where $A = \{x_1, \dots, x_n\}$.

VC dimension

- Let us define

$$m_{\mathcal{F}}(n) := \max\{|\mathcal{F}(x_1^n)| : x_1, \dots, x_n \in \mathcal{X}\}$$

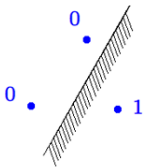
- $x_1^n = \{x_1, \dots, x_n\}$ is **shattered** by $\mathcal{F}$ if $\mathcal{F}(x_1^n) = \{0, 1\}^n$.
- I.e., shattered if restriction of $\mathcal{F}$ to x_1^n gives all possible binary functions.
- The **VC dimension** of $\mathcal{F}$ is defined as

$$\begin{aligned}\nu(\mathcal{F}) &= \sup\{n : m_{\mathcal{F}}(n) = 2^n\} \\ &= \sup\{n : \text{some } \{x_1, \dots, x_n\} \subset \mathcal{X} \text{ is shattered by } \mathcal{F}\}\end{aligned}$$

- The largest cardinality of a subset of $\mathcal{X}$ shattered by $\mathcal{F}$.
- Equivalence between binary function and sets: $\mathcal{F} = \{1_C : C \in \mathcal{C}\}$.
- Can talk about VC dimension of collections of subsets of $\mathcal{X}$.
- $\mathcal{F}$ (or $\mathcal{C}$) is called VC class if its VC dimension is finite.
- VC dimension is form of combinatorial complexity.

Examples

- $\mathcal{F} = \{1_{(-\infty, a]} : a \in \mathbb{R}\}$ has $\nu(\mathcal{F}) = 1$.
- $\mathcal{F} = \{1_{[a, b]} : a, b \in \mathbb{R}, a \leq b\}$ has $\nu(\mathcal{F}) = 2$.
- $\mathcal{F} =$ indicators of half planes in $\mathbb{R}^2$ has $\nu(\mathcal{F}) = 3$.
 - Two possible arrangement in general position, none can be shattered. See picture ⁴ What if not in general position.



⁴From HDP book.

Class	VC dimension
Pairs on intervals $[a, b] \cup [c, d]$ in $\mathbb{R}$	4
Circles in $\mathbb{R}^2$	3
Rectangles $[a, b] \times [c, d]$ in $\mathbb{R}^2$	4
Squares $[a, b]^2$ in $\mathbb{R}^2$	3
Rectangles not necessarily axis-aligned in $\mathbb{R}^2$	7
Polygons with k vertices in $\mathbb{R}^2$	$2k + 1$
Half-spaces in $\mathbb{R}^d$	$d + 1$

- Equivalence between binary function and sets: $\mathcal{F} = \{1_C : C \in \mathcal{C}\}$.
- $\mathcal{F}$ (or $\mathcal{C}$) is called VC class if its VC dimension is finite.

Theorem 15

Let $\mathcal{F}$ be nonempty, VC class with, $\nu(\mathcal{F}) = d < \infty$, then

$$m_{\mathcal{F}}(n) \leq \sum_{i=0}^d \binom{n}{i}$$

- There are only two possibilities

$$m_{\mathcal{F}}(n) \begin{cases} = 2^n & n \leq d, \\ \leq (en/d)^d & n > d \end{cases}$$

- The inequality $\sum_{i=0}^d \binom{n}{d} \leq (en/d)^d$ for $d \leq n$:

$$\left(\frac{d}{n}\right)^d \sum_{i=0}^d \binom{n}{d} \leq \sum_{i=0}^d \binom{n}{d} \left(\frac{d}{n}\right)^i \leq \left(1 + \frac{d}{n}\right)^n \leq e^d$$

using $1 + x \leq e^x$, hence $(1 + x)^n \leq e^{nx}$.

- Other bounds: $\sum_{i=0}^d \binom{n}{d} \leq (n+1)^d$.

- Equivalence between binary function and sets: $\mathcal{F} = \{1_C : C \in \mathcal{C}\}$.
- $\mathcal{F}$ (or $\mathcal{C}$) is called VC class if its VC dimension is finite.

Theorem 16

Let $\mathcal{C}$ be a nonempty, VC class collection of subsets of $\mathcal{X}$, with $\nu(\mathcal{C}) = d < \infty$. Then for any finite subset $A \subset \mathcal{X}$,

$$m_{\mathcal{C}}(A) \leq |\{B \subset A : |B| \leq d\}| = \sum_{i=0}^d \binom{|A|}{i}$$

where $m_{\mathcal{C}}(A)$ is the cardinality of the trace of $\mathcal{C}$ on A .

- Follows from Pajor's lemma: $m_{\mathcal{C}}(A) \leq |\{B \subset A : B \text{ is shattered by } \mathcal{C}\}|$.
- Note that A is finite here.
- $m_{\mathcal{C}}(A) = |\mathcal{F}(A)|$ where $\mathcal{F} = \{1_C : C \in \mathcal{C}\}$.
- $m_{\mathcal{C}}(A) = |\mathcal{C} \cap A| = |\{C \cap A : C \in \mathcal{C}\}|$.
- Consider the example $\mathcal{C} = \{(-\infty, a] : a \in \mathbb{R}\}$ with $\nu(\mathcal{C}) = 1$ and $A = \{x_1, x_2, x_3\}$.

- Recall

$$\mathcal{R}_n(\mathcal{F} \mid X_1^n) \leq \sqrt{\frac{2D_{\mathcal{F}}^2 \log m_{\mathcal{F}}(n)}{n}}.$$

- If $\nu(\mathcal{F}) = d < \infty$, we have

$$\mathcal{R}_n(\mathcal{F} \mid X_1^n) \leq \sqrt{\frac{2D_{\mathcal{F}}^2 d \log(en/d)}{n}}$$

- Similar bound holds for unconditional $\mathcal{R}_n(\mathcal{F})$:

$$\mathcal{R}_n(\mathcal{F}) \leq \sqrt{\frac{2\mathbb{E}D_{\mathcal{F}}^2 d \log(en/d)}{n}} \lesssim \sqrt{\frac{d \log(n/d)}{n}}.$$

assuming $\mathbb{E}D_{\mathcal{F}}^2 = O(1)$.

- (Note the application of Jensen inequality: $\mathbb{E}\sqrt{Z} \leq \sqrt{\mathbb{E}Z}$.)

Proposition 1

For any b -uniformly bounded loss function call $\ell_{\mathcal{F}}$, with VC dimension $\nu(\ell_{\mathcal{F}}) = d < \infty$, we have

$$\|\bar{R}_n\|_{\mathcal{F}} \leq 2\sqrt{\frac{2\mathbb{E}D_{\mathcal{F}}^2 d \log(en/d)}{n}} + \delta$$

with probability $1 - \exp(-n\delta^2/2b^2)$.

- Recall that

$$\mathbb{E}D_{\mathcal{F}}^2 = \mathbb{E} \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n f^2(X_i).$$

- Again log factor can be removed (by chaining/Dudley's integral).

VC dimension (continued)

- Relation between VC dimension and usual linear algebraic dimension.

Proposition 2

- Let $\mathcal{G}$ be a *vector space of functions* with dimension $\dim(\mathcal{G}) < \infty$.
- Let $\mathcal{S}(\mathcal{G})$ be the subgraph class of $\mathcal{G}$: $\mathcal{S}(\mathcal{G}) = \{\text{lev}_{\leq 0}(g) : g \in \mathcal{G}\}$. Then,

$$\nu(\mathcal{S}(\mathcal{G})) \leq \dim(\mathcal{G})$$

- $\text{lev}_{\leq 0}(g) = \{x : g(x) \leq 0\}$.
- If $\mathcal{S}(\mathcal{G})$ shatters $\{x_1, \dots, x_n\}$, then training a ERM classifier from the class $x \mapsto 1\{g(x) \leq 0\}$, $g \in \mathcal{G}$, on data $\{(x_1, y_1), \dots, (x_n, y_n)\}$ gives a training error of zero

Proof of Proposition 2

- Need to show no subset of $n := \dim(\mathcal{G}) + 1$ can be shattered.
- Pick $\{x_1, \dots, x_n\} \subset \mathcal{X}$, and define the (linear) map

$$\Phi : \mathcal{G} \rightarrow \mathbb{R}^n, \quad \Phi(g) = (g(x_1), \dots, g(x_n))$$

- Let $\Phi(\mathcal{G})$ denote the range of $\mathcal{G}$.
- We have $\dim(\Phi(\mathcal{G})) \leq \dim(\mathcal{G}) = n - 1 < n$.
- Hence, $\exists v \in \mathbb{R}^n \setminus \{0\}$ s.t. $\langle v, \Phi(g) \rangle = 0$ for all $g \in \mathcal{G}$.
- Assume that $\mathcal{S}(\mathcal{G})$ shatters $\{x_1, \dots, x_n\}$.
- Then, $\exists g \in \mathcal{G}$ such that (using convention $\text{sign}(0) = -1$)

$$\text{sign}(g(x_i)) = \text{sign}(v_i), \quad \forall i \in [n].$$

- WLOG, at least some $v_j > 0$ so that $\text{sign}(v_j) = 1$ and $g(x_j) > 0$.
- For that g we have $0 = \langle v, \Phi(g) \rangle = \sum_i |v_i| |g(x_i)| \geq |v_j| |g(x_j)|$, contradicting $v_j \neq 0$.

Example (half spaces)

- Consider the class of all half-spaces in $\mathbb{R}^d$:

$$\mathcal{C} = \{ \{x \in \mathbb{R}^d : \langle a, x \rangle + b \leq 0\} : (a, b) \in \mathbb{R}^d \times \mathbb{R} \}$$

- $\mathcal{C}$ is sub-graph class for $\{x \mapsto \langle a, x \rangle + b : a \in \mathbb{R}^d, b \in \mathbb{R}\}$.
- $\nu(\mathcal{C}) \leq d + 1$.
- (VC dimension is $= d + 1$ in fact)

Example (spheres)

- Consider spheres in $\mathbb{R}^d$:

$$\mathcal{C} = \{ \{x \in \mathbb{R}^d : \|x - a\| \leq b\} : (a, b) \in \mathbb{R}^d \times \mathbb{R}_+ \}$$

- $\mathcal{C}$ is sub-graph class for functions:

$$\mathcal{F}_1 := \left\{ x \mapsto \|x\|^2 - 2 \sum_{j=1}^d a_j x_j + \|a\|^2 - b^2 : (a, b) \in \mathbb{R}^d \times \mathbb{R}_+ \right\}$$

- Define $\phi(x) := (1, x, \dots, x_d, \|x\|^2)$, maps $\mathbb{R}^d \rightarrow \mathbb{R}^{d+2}$.
- Class $\mathcal{F}_1$ is included in

$$\mathcal{F}_2 = \{ x \mapsto \langle \theta, \phi(x) \rangle : \theta \in \mathbb{R}^{d+2} \}, \quad \dim(\mathcal{F}_2) = d + 2.$$

- Hence, $\nu(\mathcal{C}) \leq d + 2$.
- (Can show that VC dimension = $d + 1$.)

Neural Network (NN) examples

- Linear threshold function (t-lin.):

$$\mathcal{F}_d^{\text{LT}} = \{x \mapsto \text{sign}(\langle \theta, x \rangle + \theta_0) : \theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}\}.$$

has $\nu(\mathcal{F}_d^{\text{LT}}) = d + 1$.

- Also let us introduce

$$\mathcal{F}^{\text{LT}}(\Theta) = \{x \mapsto \text{sign}(\langle \theta, x \rangle + \theta_0) : \theta \in \Theta, \theta_0 \in \mathbb{R}\}$$

- Also recall

$$\mathcal{R}_n(\mathcal{F}) \leq \sqrt{\frac{2\mathbb{E}D_{\mathcal{F}}^2 \log m_{\mathcal{F}}(n)}{n}}.$$

Two-layer network

- Two-layer network with linear threshold functions in 1st layer, ℓ_1 bounded weights in 2nd layer:

$$\mathcal{F}_d^{(1)} := \left\{ \sum_{i=1}^k \theta_i f_i : k \in \mathbb{N}, f_i \in \mathcal{F}_d^{\text{LT}}, \|\theta\|_1 \leq 1 \right\} = \text{co}(\mathcal{F}_d \cup -\mathcal{F}_d)$$

since $0 \in \mathcal{F}_d$.

- Also called **absolute convex hull** or symmetric convex hull.

$$\text{absconv}(A) = \left\{ \sum_{i=1}^k \lambda_i x_i : k \in \mathbb{N}, x_i \in A, \sum_{i=1}^k |\lambda_i| \leq 1 \right\}$$

- $\mathcal{F}_d^{(1)} := \text{absconv}(\mathcal{F}_d^{\text{LT}})$, hence

$$\mathcal{R}_n(\mathcal{F}_d^{(1)}) = \mathcal{R}_n(\mathcal{F}_d^{\text{LT}}) \lesssim \sqrt{\frac{d \log(en/d)}{n}}$$

Why?

- Recalling Rademacher complexity of the function class $\mathcal{F}$:

$$\mathcal{R}_n(\mathcal{F}) = \mathbb{E}_{X, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \right]. \quad (14)$$

- Rademacher complexity is invariant under taking convex hull:

$$\mathcal{R}_n(\text{co}(\mathcal{F})) = \mathcal{R}_n(\mathcal{F})$$

(Hint: maximizing a convex function over a domain.)

- $\text{absconv}(\mathcal{F}) = \text{co}(\{0\} \cup \mathcal{F} \cup -\mathcal{F})$.
- Hence, $\mathcal{R}_n(\text{absconv } \mathcal{F}) = \mathcal{R}_n(\mathcal{F})$.

Two-layer network

- Two-layer network with **bounded fan-in** linear threshold functions in first layer and ℓ_1 bounded weights in second layer:

$$\mathcal{F}_{d,s} = \{x \mapsto \text{sign}(\langle \theta, x \rangle + \theta_0) : \theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R}, \|\theta\|_0 \leq s\}$$

$$\mathcal{F}_{d,s}^{(1)} = \text{absconv}(\mathcal{F}_{d,s})$$

- We have $\mathcal{F}_{d,s} = \mathcal{F}^{\text{LT}}(\mathbb{B}_0(s)) = \bigcup_{|S|=k} \mathcal{F}_S$ where

$$\mathcal{F}_S = \mathcal{F}^{\text{LT}}(\{\theta \in \mathbb{R}^d : \theta_{S^c} = 0\}), \quad m_{\mathcal{F}_S}(n) \leq \left(\frac{en}{s+1}\right)^{s+1}$$

- Since $m_{\mathcal{F} \cup \mathcal{G}}(n) \leq m_{\mathcal{F}}(n) + m_{\mathcal{G}}(n)$,

$$m_{\mathcal{F}_{d,s}}(n) \leq \left(\frac{en}{s+1}\right)^{s+1} \binom{d}{s} \implies \mathcal{R}(\mathcal{F}) \lesssim \sqrt{\frac{s \log(nd/s)}{n}}$$

Network of LT functions

Theorem 17

Let $\mathcal{F}_{p,k}$ be the class of functions computed by a feed-forward network of LT functions with k computation units and p (total) parameters. Then,

$$m_{\mathcal{F}_{p,k}}(n) \leq \left(\frac{enk}{p}\right)^p$$

and hence $\nu(\mathcal{F}_{p,k}) \leq 2p \log_2(2k/\log 2)$.

- That is, $\nu(\mathcal{F}_{p,k}) = O(p \log k)$.
- Dimension of the input space $\mathcal{X} \subset \mathbb{R}^d$ does not come into play.

Proof

- Fix a set of n input vector $x_1, \dots, x_n$.
- Consider a topological sort of computation units. (NN architecture is a DAG.)
- p_ℓ : number of parameters of unit ℓ .
- D_ℓ : number of distinct **states** of unit ℓ .
(# of distinct functions $\{x_1, \dots, x_n\} \rightarrow \{\pm 1\}^\ell$ from input to the output of units before ℓ , as we vary the parameters. See next slide.)
- $D_1 \leq (en/p_1)^{p_1}$, and $D_\ell \leq D_{\ell-1}(en/p_\ell)^{p_\ell}$, hence

$$D_k \leq \prod_{\ell=1}^k (en/p_\ell)^{p_\ell}$$

- Since the bound is independent of x_1^n , we have

$$\log m_{\mathcal{F}_{p,k}}(n) \leq \sum_{\ell=1}^k p_\ell \log(en/p_\ell)$$

- Bound is maximized by spreading p uniformly over the k units: $p_\ell = p/k$.

Details:

- To see what is going on with *states*, consider

$$f(g_1(x), g_2(x), \dots, g_\ell(x))$$

all binary functions.

- Consider how many function on $\{x_1, \dots, x_n\}$ we have as g_i and f vary

$$x \mapsto f(g_1(x), g_2(x), \dots, g_\ell(x)), \quad \forall f, g_1, \dots, g_\ell$$

- First consider functions $\{x_1, \dots, x_n\} \rightarrow \{\pm 1\}^\ell$ generated by

$$(g_1(x), g_2(x), \dots, g_\ell(x)), \quad \forall g_1, \dots, g_\ell$$

- Each such function can be represented by a binary matrix: ($n = 4, \ell = 2$)

	1	2		1	2				1	2	
x_1	+	-		x_1	-	-			x_1	+	+
x_2	-	+		x_2	+	-	...		x_2	+	-
x_3	+	+		x_3	+	-			x_3	-	-
x_4	-	-		x_4	-	+			x_4	-	+

- First consider functions $\{x_1, \dots, x_n\} \rightarrow \{\pm 1\}^\ell$ generated by

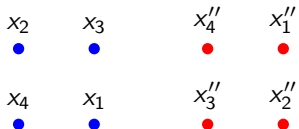
$$(g_1(x), g_2(x), \dots, g_\ell(x)), \quad \forall g_1, \dots, g_\ell$$

- Each such function can be represented by a binary matrix:

	1	2		1	2			1	2
x_1	+	-	x'_1	-	-		x''_1	+	+
x_2	-	+	x'_2	+	-	...	x''_2	+	-
x_3	+	+	x'_3	+	-		x''_3	-	-
x_4	-	-	x'_4	-	+		x''_4	-	+

- For each matrix, we have possibly different set of n input vectors to the function f . (Each row is a data point to f .)
- Each of these matrices produces at most $(en/p_\ell)^{p_\ell}$ functions $\{x_1, \dots, x_n\} \rightarrow \{\pm 1\}$ at the output of f , and
- the number of matrices is D_ℓ .

- I.e, the same set $\{x_1, x_2, x_3, x_4\}$ at the input layer, *appear* as different configurations at the input to f , due to the variations in $g_1, \dots, g_\ell$.



$\{x_1, x_2\}$ cannot be separated from $\{x_3, x_4\}$ in blue configuration, but can be in red.

Network of LT functions (lower bound)

Theorem 18

Let $\mathcal{F}_{d,k}$ be the class of functions $f : \mathbb{R}^d \rightarrow \{\pm 1\}$ computed by a two-layer feed-forward network of LT functions with k hidden computation units (i.e., $p = k(d + 2) + 1$ parameters). Then $\nu(\mathcal{F}_{d,k}) \geq kd = \Omega(p)$.

- More involved argument shows $\nu(\mathcal{F}_{d,k}) = \Omega(p \log k)$.
- $p = k(d + 1) + (k + 1)$.
- Conclusion: For LT functions with p parameters and k computation unit, VC dimension is $\Theta(p \log k)$ independent of depth.

Proof sketch

- Arrange kd points in k well-separated clusters $C_i, i = 1, \dots, k$ on the surface of a sphere in $\mathbb{R}^d$.
- Ensure d points in each cluster are in general position.
- Choose the decision boundary of i th hidden neuron to pass through point in C_i , and orient to have output at the center of sphere.
- Choose parameters of the output unit to compute conjunction of its k inputs.
- By perturbing hidden unit parameters all 2^{kd} classifications can be computed.

Other nonlinearities

- Linear threshold networks are piecewise constant.
- What about piecewise linear, networks of ReLU $x \mapsto (x)_+$? Or piecewise polynomials?
- How about smooth nonlinearities: sigmoid, soft-max, smoothed ReLU

$$\sigma(t) = \frac{1}{1 + e^{-t}}, \quad [\sigma(t)] = \frac{e^{t_i}}{\sum_j e^{t_j}}, \quad \sigma(t) = \log(1 + e^t)$$

not clear if finite.

- A smooth function with few parameters can have infinite VC dimension.

Sine nonlinearity

- Consider $\mathcal{F}_{\sin} = \{x \mapsto \text{sign}(\sin(\theta x)) : \theta \in \mathbb{R}\}$.
- We have $\nu(\mathcal{F}_{\sin}) = \infty$.
- Any sequence $x_i = 2^i, i = 1, 2, \dots, n$ can be shattered.
- Consider the subclass obtained with $\theta = c\pi$ where c has binary representation $0.b_1b_2\dots b_n1$ for $b_i \in \{0, 1\}$. Then,

$$\begin{aligned}\sin(x_i\theta) &= \sin(2^i\pi 0.b_1b_2\dots b_n1) \\ &= \sin(\pi b_1b_2\dots b_i.b_{i+1}\dots b_n1) \\ &= \sin(\pi b_i.b_{i+1}\dots b_n1)\end{aligned}$$

so that $\text{sign}(\sin(\theta x_i)) = 1 - 2b_i$.

Geometric methods

- Want to bound growth function $m_{\mathcal{F}}(n)$ of a parametric family:

$$\{x \mapsto f(x, \theta) : \theta \in \Theta\}, \quad \Theta \subset \mathbb{R}^p$$

- Example: linear threshold functions $f(x, \theta) = \text{sign}(\langle \theta, x \rangle)$:
- Intercept included in $\theta \in \mathbb{R}^d$ and x has been augmented with 1:

$$x = (1, x_1, \dots, x_n), \quad \theta = (\theta_0, \theta_1, \dots, \theta_d)$$

- We know $\nu(\mathcal{F}_d^{\text{LT}}) = d + 1$, hence $m_{\mathcal{F}_d^{\text{LT}}}(n) \leq \sum_{i=0}^{d+1} \binom{n}{i}$.
- Directly compute $m_{\mathcal{F}_d^{\text{LT}}}$ using a geometric argument.
- Useful in other situations.

Theorem 19

For the class of linear threshold functions on $\mathbb{R}^d$,

$$m_{\mathcal{F}}(n) = 2 \sum_{i=0}^d \binom{n-1}{i}$$

Proof idea:

- Fix $x_1, \dots, x_n \in \mathbb{R}^{d+1}$.
- Divide the parameter space into cells that give the same classification.
- Count the cells using a geometric argument (goes back to Schaffli, 1851)

- Assume the points are in $\mathbb{R}^d$ (for simplicity) and
- in **general position**, that is every subset of size $\leq d$ is linearly independent.
- For each x_i define the hyperplane

$$H_{x_i} = \{\theta \in \mathbb{R}^d : \langle \theta, x_i \rangle = 0\}$$

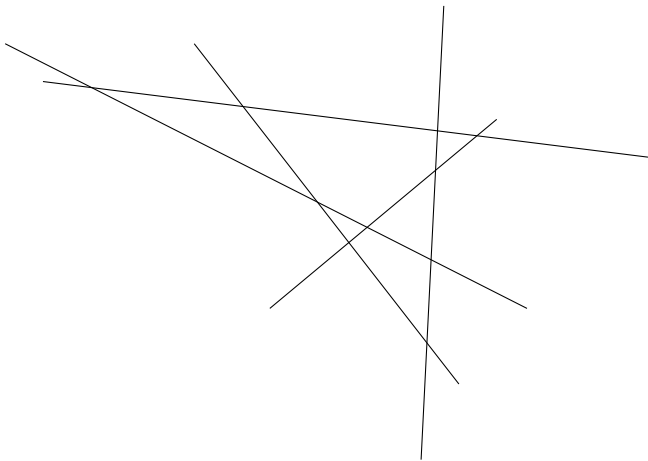
- Number of classifications on x_1^n is given by

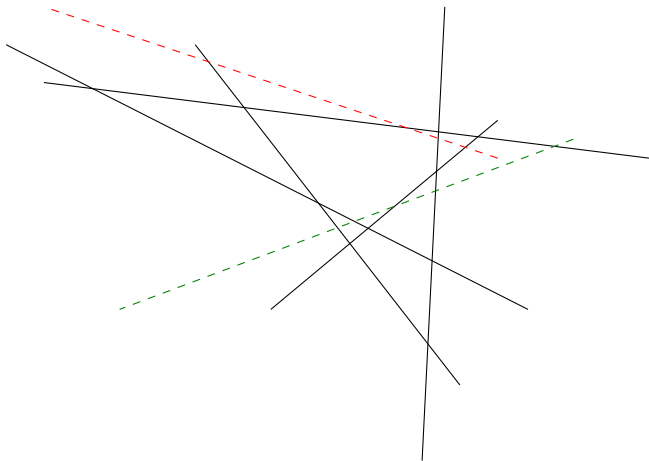
$$|\mathcal{F}(x_1^n)| = \left| \text{CC} \left(\mathbb{R}^d \setminus \bigcup_{i=1}^n H_{x_i} \right) \right| =: C(n, d)$$

- Compute $C(n, d)$ by induction on n (also shows independence of x_1^n).
- $C(1, d) = 2$.
- Consider arrangement by n planes, and add new point x_{n+1} .
- $H_{x_{n+1}}$ splits some of the cells, leaves others intact.
- Number of cells split = $|\text{CC}(H_{x_{n+1}} \setminus \bigcup_{i=1}^n H_{x_i})| = C(n, d-1)$, thus

$$C(n+1, d) = C(n, d) + C(n, d-1).$$

- Solution to this is $C(n, d) = 2 \sum_{k=0}^{d-1} \binom{n-1}{k}$.





Convex losses for classification

- Finding ERM for zero-loss is difficult.
- Let $\mathcal{Y} \in \{-1, 1\}$, $\mathcal{Y}' = \mathbb{R}$, i.e. $f : \mathcal{X} \rightarrow \mathbb{R}$.
- Looking at general loss functions of the form $\ell(y, t) = \phi(yt)$.
- Risk for zero-loss is

$$\begin{aligned} R(f) &= \mathbb{P}[Y \neq \text{sign}(f(X))] = \mathbb{E}1\{Yf(X) \leq 0\} \\ &=: \mathbb{E}\phi_0(Yf(X) \leq 0) \end{aligned}$$

where $\phi_0(t) := 1\{t \leq 0\}$.

- Can replace ϕ_0 with a convex loss function:

$$\phi_{\text{svm}}(t) = (1 - t)_+, \quad \phi_{\text{ada}}(t) = e^{-t}, \quad \phi_{\text{logi}}(t) = \log(1 + e^{-t})$$

- Efficient convex optimization algorithms.

- What do we lose? Focus on the population level question.
- Does minimizing $R_\phi(f) = \mathbb{E}\phi(Yf(X))$ give good classifier,
- i.e., small zero-one risk $R(f) = \mathbb{E}\phi_0(Yf(X))$?
- If $\phi(t) \leq c\phi_0(t) \implies R_\phi(f) \leq c R_{\phi_0}(f)$.
- Can we do better?
- Can approximately minimizing $\mathcal{E}_\phi(f)$, also approximately minimize $\mathcal{E}_{\phi_0}(f)$?

- Posterior risk

$$C_\eta(t) = \mathbb{E}_\eta[L(Y, t)], \quad C_\eta^* = \inf_{t \in \mathbb{R}} C(t; \eta) \quad (15)$$

where η is a distribution on Y , here identified with $\eta = \mathbb{P}(Y = 1)$.

- Later we will with $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$.
- Here L is the zero one loss.
- $\tilde{C}$ and $\tilde{C}^*$ corresponding quantities for a surrogate loss $\tilde{L}$.
- Recall $C_{\eta(x)}(t) = \mathbb{E}[L(Y, t) \mid X = x]$. Hence,

$$\mathbb{E}C_{\eta(X)}(f(X)) = \mathbb{E}L(Y, f(X)) = R(f).$$

For margin-based classifiers $L(y, t) = \phi(yt)$

$$C_\eta(t) = \eta\phi(t) + (1 - \eta)\phi(-t)$$

Calibration

- Often drop dependence on η for simplicity.
- ε -approximate minimizers of $C(t)$:

$$\mathcal{M}(\varepsilon) = \mathcal{M}_\eta(\varepsilon) = \{t \in \mathbb{R} : C(t) - C^* < \varepsilon\}$$

- *calibration function*

$$\delta(\varepsilon) := \inf_{t \notin \mathcal{M}(\varepsilon)} \tilde{C}(t) - \tilde{C}^*$$

- convention $\delta(\varepsilon) = \infty$ if $C^* = \infty$. Note that

$$C(t) - C^* \geq \varepsilon \implies \tilde{C}(t) - \tilde{C}^* \geq \delta(\varepsilon), \quad \forall t \in \mathbb{R}, \varepsilon > 0$$

- For any fixed $t \in \mathbb{R}$, taking the supremum over $\varepsilon \leq C(t) - C^*$ and using the fact that $\varepsilon \mapsto \delta(\varepsilon)$ is increasing we obtain

$$\tilde{C}(t) - \tilde{C}^* \geq \sup_{\varepsilon \leq C(t) - C^*} \delta(\varepsilon) = \delta(C(t) - C^*)$$

assuming that $\delta(\cdot)$ is continuous.

Calibration

- Putting η back in, we have shown

$$\delta_{\eta}(C_{\eta}(t) - C_{\eta}^*) \leq \tilde{C}_{\eta}(t) - \tilde{C}_{\eta}^*.$$

Definition 1 (Calibration)

- $\tilde{L}$ is calibrated for L (w.r.t. $\mathcal{Q}$) if

$$\delta_{\eta}(\varepsilon) > 0, \quad \forall \varepsilon > 0, \eta \in \mathcal{Q}$$

- $\tilde{L}$ is uniformly calibrated for L (w.r.t. $\mathcal{Q}$) if

$$\delta_{\eta}(\varepsilon) \geq \delta(\varepsilon) > 0, \quad \forall \varepsilon > 0, \eta \in \mathcal{Q}$$

- Usually define $\delta(\varepsilon) := \delta_{\mathcal{Q}}(\varepsilon) := \inf_{\eta \in \mathcal{Q}} \delta_{\eta}(\varepsilon)$
- but any uniform lower bound suffices.

- Let $B_f := \sup_x |C_{\eta(x)}(f(x))|$.
- $\mathbb{P}$ on $\mathcal{X} \times \mathcal{Y}$ is of $\mathcal{Q}$ -type if $\mathbb{P}(Y \in \cdot | X = x)$ belongs to $\mathcal{Q}$.

Theorem 20

- $\tilde{L}$ is uniformly calibrated for L with calibration function $\delta(\cdot)$.
- $\delta^{**} : [0, B_f] \rightarrow [0, \infty]$ be the Fenchel-Legendre biconjugate of $\delta|_{[0, B_f]}$.
- Assume $R^*(f)$ and $\tilde{R}^*(f)$ are finite.

Then,

$$\delta_{B_f}^{**}(R(f) - R^*) \leq \tilde{R}(f) - \tilde{R}^*$$

- For any proper function f , its biconjugate $f^{**} \leq f$ and
- f^{**} is convex and lsc (lower semicontinuous).
- In fact, f^{**} is the largest convex lsc minorant of f .
- For a proper function $f = f^{**}$ iff f is convex lsc.
- Alternatively, $\text{epi}(f^{**}) = \text{closed convex hull of } \text{epi}(f)$.

Example 4

- For classification $L(y, t) = 1\{yt \leq 0\} = 1\{y \operatorname{sign}(t) \leq 0\}$. Hence,

$$C_\eta(t) = \eta 1\{t \leq 0\} + (1 - \eta) 1\{t \geq 0\}$$

- We have $C_\eta^* = \min\{\eta, 1 - \eta\}$. Hence,

$$\begin{aligned} C_\eta(t) - C_\eta^* &= \max\{0, 2\eta - 1\} 1\{t \leq 0\} + \max\{1 - 2\eta, 0\} 1\{t \geq 0\} \\ &= |2\eta - 1| 1\{(2\eta - 1)t \leq 0\}. \end{aligned}$$

- Then,

$$\mathcal{M}_\eta(\varepsilon) = \begin{cases} \mathbb{R} & \varepsilon > |2\eta - 1| \\ \{t \in \mathbb{R} : (2\eta - 1)t > 0\} & \varepsilon \leq |2\eta - 1| \end{cases}$$

- It follows that $(\inf \emptyset = \infty)$

$$\delta_\eta(\varepsilon) = \begin{cases} \infty & \varepsilon > |2\eta - 1| \\ \inf_t \{\tilde{C}_\eta(t) - \tilde{C}_\eta^* : (2\eta - 1)t \leq 0\} & \varepsilon \leq |2\eta - 1| \end{cases}$$

Proof

- Let $Z := C_{\eta(X)}(f(X)) - C_{\eta(X)}^*$,
- and $\tilde{Z} := \tilde{C}_{\eta(X)}(f(X)) - \tilde{C}_{\eta(X)}^*$, both random variables.
- By assumption $|Z| \leq B_f$. We have

$$\delta^{**}(Z) \leq \delta(Z) \leq \delta_{\eta(X)}(Z) \leq \tilde{Z}.$$

- Since δ^{**} is convex, we can apply Jensen inequality.

$$\delta^{**}(\mathbb{E}Z) \leq \mathbb{E}\delta^{**}(Z) \leq \mathbb{E}\tilde{Z}.$$

- $\mathbb{E}Z$ and $\mathbb{E}\tilde{Z}$ are the excess risk of the target and surrogate losses.

Lemma 2

For margin-based loss $\tilde{L}$ we define

$$H(\eta) = \inf_{t \in \mathbb{R}: (2\eta-1)t \leq 0} \tilde{C}_\eta(t) - \tilde{C}_\eta^* \quad (16)$$

1. $\tilde{L}$ *classif. calibrated* iff $H(\eta) > 0$ for all $\eta \neq 1/2$.
2. If $\tilde{L}$ is continuous, then $\delta_\eta(\varepsilon) = H(\eta)$ for all $\varepsilon \leq |2\eta - 1|$.
3. H is continuous, $H(\eta) = H(1 - \eta)$ and $H(1/2) = 0$.

Theorem 21

- For margin-based loss $\tilde{L}$, the following are equivalent:
 1. $\tilde{L}$ is classification calibrated.
 2. $\tilde{L}$ is uniformly classification calibrated.
- Moreover for H as in (16), let

$$\psi(\varepsilon) = H\left(\frac{1+\varepsilon}{2}\right), \quad \varepsilon \in [0, 1].$$

- Then, $\psi^{**}(\varepsilon) \leq \delta_{\mathcal{Q}}^{**}(\varepsilon)$ for all $\varepsilon \in [0, 1]$, with equality if $\tilde{L}$ is continuous.
- If $\tilde{L}$ is classification calibrated, we have $\psi^{**}(\varepsilon) > 0$ for all $\varepsilon > 0$.

- Conclusion

$$\psi^{**}(R(f) - R^*) \leq \tilde{R}(f) - \tilde{R}^*$$

Proof

- We have

$$\delta_Q(\varepsilon) = \inf_{\eta \in [0,1]} \delta_\eta(\varepsilon) \stackrel{\geq}{=} \inf_{|2\eta-1| \geq \varepsilon} H(\eta) = \inf_{\eta \geq (1+\varepsilon)/2} H(\eta)$$

by symmetry of H around $1/2$. (The second one could be an inequality due to the issue with $t = 0$. Assuming $\tilde{L}$ is continuous hence $\tilde{C}(t)$ is continuous at $t = 0$ that won't happen.)

- Assuming $\tilde{L}$ is calibrated, H is strictly positive on $[1 + \varepsilon/2, 1]$,
- and continuous (by previous lemma),
- hence $\delta_Q^{**}(\varepsilon) > 0$, i.e., uniformly calibrated.
- Now, we have $\delta_Q^{**}(\cdot) = \psi^{**}$,
- since $\delta_Q(\varepsilon) = \inf_{\varepsilon' \geq \varepsilon} \psi(\varepsilon')$.

Theorem 22

$\tilde{L}$ margin-based convex loss represented by ϕ .

- (i) $\tilde{L}$ is classification calibrated.
- (ii) ϕ is differentiable at 0 and $\phi'(0) < 0$.

If $\tilde{L}$ is classification calibrated, then $\delta_{\mathcal{Q}}^{**}(\varepsilon) = \phi(0) - \tilde{C}_{(\varepsilon+1)/2}^*$.

Proof

- **bad notation** for $\tilde{C}^*$, this is not a conjugate
- (ii) $\implies$ (i): ϕ differentiable at 0 implies $\tilde{C}_\eta(\cdot)$ is differentiable at 0,
- with $\tilde{C}'_\eta(0) = (2\eta - 1)\phi'(0) < 0$ for $\eta \in (1/2, 1]$.
- Convexity $\tilde{C}_\eta(\cdot)$ implies is decreasing on $(-\infty, 0]$, hence

$$\begin{aligned} H(\eta) &= \inf_{t \in \mathbb{R}: (2\eta-1)t \leq 0} \tilde{C}_\eta(t) - \tilde{C}_\eta^* \\ &= \inf_{t \leq 0} \tilde{C}_\eta(t) - \tilde{C}_\eta^* \\ &= \tilde{C}_\eta(0) - \tilde{C}_\eta^* = \phi(0) - \tilde{C}_\eta^* \end{aligned}$$

for $\eta \in (1/2, 1]$.

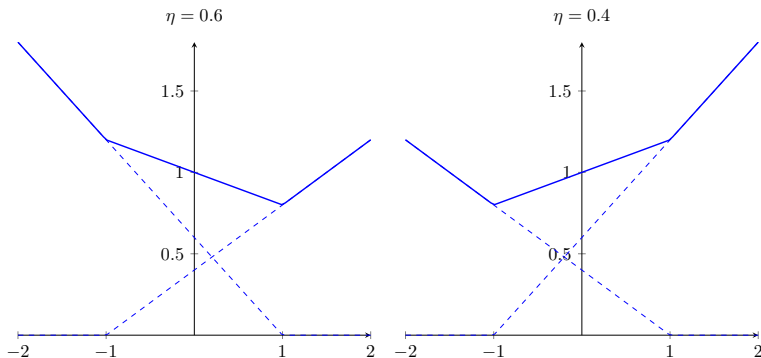
- $\tilde{C}_\eta^*$ is the minimizer of $\tilde{C}_\eta(\cdot)$ over $\mathbb{R}$.
- Since $\tilde{C}'_\eta(0) < 0$ implies that $\tilde{C}_\eta$ cannot have a minimizer at 0,
- $\tilde{C}_\eta(0) - \tilde{C}_\eta^* > 0 \implies H(\eta) > 0$ showing classif. calibration.

- (i) $\implies$ (ii): Assume ϕ is diff'able at 0 (not needed). Assume $\phi'(0) \geq 0$ and derive contradiction:

$$\tilde{C}_1(t) = \phi(t) \geq \phi'(0)t + \phi(0) \geq \tilde{C}_1(0), \quad \forall t > 0. \quad (17)$$

- $\tilde{L}$ is classif. calib. $\implies H(\eta) > 0 \implies \tilde{C}_\eta^* = \inf_{t>0} \tilde{C}_\eta(t)$ for $\eta > 1/2$.
- Apply with $\eta = 1$, take inf over $t > 0$ in (17) $\implies \tilde{C}_1^* \geq \tilde{C}_1(0)$,
- showing $H(1) \leq 0$ contradicting classif. calibration.
- Last part: $\tilde{C}'_{1/2}(0) = \frac{1}{2}\phi'(0) - \frac{1}{2}\phi'(0) = 0$ hence $\tilde{C}_{1/2}^* = \tilde{C}_{1/2}(0)$.
- Shows $H(\eta) = \phi(0) - \tilde{C}_\eta^*$ holds even at $\eta = 1/2$, i.e. for all $\eta \in [1/2, 1]$.
- $\tilde{C}_\eta^*$ infimum of affine functions $\implies$ concave,
- $\implies H$ is convex on $[1/2, 1]$.
- H is continuous (follows from continuity of $\tilde{L}$?) and convex,
- $\implies H = H^{**}$.
- $\psi = \psi^{**}$ as well. (Recall $\psi(\varepsilon) = H(\frac{1+\varepsilon}{2})$.)

Example



- $\phi(t) = (1 - t)_+ = \max(0, 1 - t)$, hinge loss.
- $\tilde{C}_\eta(t) = \eta(1 - t)_+ + (1 - \eta)(1 + t)_+$.
- $\tilde{C}_\eta^* = \tilde{C}_\eta(1) = 2(1 - \eta)$ if $\eta > 1/2$ and $= \tilde{C}_\eta(-1) = 2\eta$ if $\eta < 1/2$

- $H(\eta) = 1 - \tilde{C}_\eta^* = 1 - 2 \min(\eta, 1 - \eta)$.
- For $\varepsilon \in [0, 1]$:

$$\psi(\varepsilon) = H\left(\frac{1+\varepsilon}{2}\right) = 1 - 2 \min\left(\frac{1+\varepsilon}{2}, \frac{1-\varepsilon}{2}\right) = 1 - 2\frac{1-\varepsilon}{2} = \varepsilon.$$

- Clear that $\psi^{**} = \psi$. Conclusion:

$$R(f) - R^*(f) \leq R_{\text{hinge}}(f) - R_{\text{hinge}}^*.$$

- Excess risk in 0-1 loss is directly bounded by excess risk in hinge loss.
- Also, called Zhang's inequality.

Hard-margin SVM

- Linearly separable data.
- Distance of x_i to the hyperplane $\{x : w^T x + b = 0\}$ is $(w^T x_i + b)/\|w\|$.
- Maximum margin separating hyperplane:

$$\max_{w,b} \min_i \frac{w^T x_i + b}{\|w\|}.$$

- Equivalent to

$$\begin{aligned} \max_{\gamma, w} \quad & \gamma / \|w\| \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq \gamma, \quad \forall i = 1, \dots, n. \end{aligned}$$

- Invariant to scaling of w, b and γ . So set $\gamma = 1$:

$$\begin{aligned} \min_w \quad & \|w\| \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq 1, \quad \forall i = 1, \dots, n, \end{aligned}$$

which is a convex problem.

Soft-margin SVM

- What if data is not linearly separable?
- Introduce slack variables $y_i w^T x_i + b \geq 1 - \xi_i$, and $\xi_i \geq 0$.
- But also need ξ_i not to be too large; can penalized them in the objective:

$$\begin{aligned} \min_{w, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y_i (w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, n \end{aligned} \tag{18}$$

- SVM penalizes data lying in the margin in addition to those misclassified.

- Set $b = 0$ for simplicity.
- Conditions $y_i w^T x_i \geq 1 - \xi_i$ and $\xi_i \geq 0$ are equivalent to

$$\xi_i \geq \max\{0, 1 - y_i w^T x_i\}$$

- Optimizing over ξ_i first, we obtain equivalent form of (18):

$$\min_w \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \max\{0, 1 - y_i w^T x_i\}$$

- Defining the **hinge loss** $L_{\text{hinge}}(y, t) = (1 - yt)_+ = \max\{0, 1 - yt\}$:

$$\min_w \frac{1}{n} \sum_{i=1}^n L_{\text{hinge}}(y_i, w^T x_i) + \lambda \|w\|^2$$

where $\lambda = 1/(2nC)$.

SVM dual

$$\begin{aligned} \min_{w, \xi, b} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, n \end{aligned}$$

- Can write in minimax form

$$\inf_{w, \xi, b} \sup_{\alpha, \mu \geq 0} L(w, \xi, b; \alpha, \mu)$$

- where the Lagrangian is

$$\begin{aligned} L(w, \xi, b; \alpha, \mu) &= \frac{1}{2} \|w\|^2 + C \sum_i \xi_i + \sum_i \alpha_i (1 - \xi_i - y_i(w^T x_i + b)) - \sum_i \mu_i \xi_i \\ &= \frac{1}{2} \|w\|^2 - w^T \left(\sum_i x_i y_i \alpha_i \right) + b \sum_i y_i \alpha_i + \sum_i \xi_i (C - \mu_i - \alpha_i) + \sum_i \alpha_i \end{aligned}$$

- Strong duality

$$\inf_{w, \xi, b} \sup_{\alpha, \mu \geq 0} L(w, \xi, b; \alpha, \mu) = \sup_{\alpha, \mu \geq 0} \inf_{w, \xi, b} L(w, \xi, b; \alpha, \mu)$$

- Minimizing over w gives $w^* = \sum_i x_i y_i \alpha_i$.
- Minimizing over b gives $-\infty$ unless $\sum_i y_i \alpha_i = 0$.
- Minimizing over ξ gives $-\infty$ unless $C - \mu_i - \alpha_i = 0, \forall i$.
- The dual problem is

$$\begin{aligned} \sup_{\alpha, \mu \geq 0} \quad & \sum_i \alpha_i - \frac{1}{2} \left\| \sum_i x_i y_i \alpha_i \right\|^2 \\ \text{s.t.} \quad & C - \alpha_i - \mu_i = 0, \forall i, \quad \sum_i \alpha_i y_i = 0 \end{aligned}$$

- Role of μ is to only enforce $C - \alpha_i = \mu_i \geq 0$.
- Also, $\sum_i \alpha_i - \frac{1}{2} \left\| \sum_i x_i y_i \alpha_i \right\|^2 = \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$.

- The dual problem simplifies to

$$\begin{aligned} \sup_{\alpha} \quad & \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C, \quad \forall i, \quad \sum_i \alpha_i y_i = 0 \end{aligned}$$

- To solve the dual, we only need the Gram matrix $(\langle x_i, x_j \rangle) \in \mathbb{R}^{n \times n}$.

KKT conditions

- Write $f^*(x) = \langle w^*, x \rangle + b^*$. KKT conditions are

$$w^* = \sum_i \alpha_i^* x_i y_i$$

$$\alpha_i^* [1 - \xi_i - y_i f^*(x_i)] = 0$$

$$\mu_i^* \xi_i^* = 0 \iff (C - \alpha_i^*) \xi_i^* = 0$$

$$\alpha^* \in [0, C]^n, \quad \sum_i y_i \alpha_i^* = 0, \quad \xi_i^* \geq 0$$

two middle lines complementary slackness, last line primal and dual feasibility.

- Note that $f^*(x) = \sum_i \alpha_i^* y_i \langle x_i, x \rangle + b^*$.
- $\alpha_i^* > 0 \implies 1 - y_i f^*(x_i) = \xi_i^*$.
- $\alpha_i^* < C \implies \xi_i^* = 0$.
- $\alpha_i^* \in (0, C) \implies y_i f^*(x_i) = 1$.
- $y_i f^*(x_i) > 1 \implies \alpha_i^* = 0$, and $\alpha_i^* = 0 \implies y_i f^*(x_i) \geq 1$
- $\alpha_i^* = C \implies y_i f^*(x_i) \leq 1$. (Misclassified points are among these: $y_i f^*(x_i) < 0$.)

SVM: General feature map Φ

- Primal problem

$$\begin{aligned} \min_{w, \xi, b} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y_i(w^T \Phi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, n \end{aligned}$$

- Dual problem

$$\begin{aligned} \sup_{\alpha} \quad & \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle \Phi(x_i), \Phi(x_j) \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C, \quad \forall i, \quad \sum_i \alpha_i y_i = 0 \end{aligned}$$

- $f^*(x) = \langle w^*, \Phi(x) \rangle + b^* = \sum_{i=1}^n \alpha_i^* y_i \langle \Phi(x), \Phi(x_i) \rangle$.
- $K(x, y) = \langle \Phi(x), \Phi(y) \rangle$ is a kernel.
- The kernel trick: only the kernel is needed not the feature map.

RKHS

- A Hilbert space is a complete inner product space.
- complete = Cauchy sequences converge.

Theorem 23

Let $\mathcal{H}$ be Hilbert space and $L : \mathcal{H} \rightarrow \mathbb{R}$ a *linear* functional. The following are equivalent:

- (a) L is continuous.
- (b) L is continuous at 0.
- (c) L is continuous at some point.
- (d) L is *bounded*: $\exists c > 0$ s.t. $|L(h)| \leq c\|h\|$ for all $h \in \mathcal{H}$.

- Norm of a (linear) functional is defined as

$$\|L\| := \sup\{|L(h)| : \|h\| \leq 1\}$$

L is bounded iff $\|L\| < \infty$.

$$\text{dist}(h, K) := \inf\{\|h - k\| : k \in K\}$$

Theorem 24

Let $\mathcal{H}$ be Hilbert space, and $h \in \mathcal{H}$.

- $K \subset \mathcal{H}$ nonempty, closed, and convex. $\exists! k_0 \in K$ s.t. $\|h - k_0\| = \text{dist}(h, K)$.
- $M \subset \mathcal{H}$ closed, linear subspace, then $f_0 \in M$ is the unique point that $\|h - f_0\| = \text{dist}(h, M)$ iff $h - f_0 \perp M$.
- The unique point above is called the projection of h onto K or M .
- In the case of M (linear closed subspace) an orthogonal projection.

Theorem 25

$M \subset \mathcal{H}$ closed, linear subspace, $h \in \mathcal{H}$ and Ph be the projection of h onto M .

- P is a linear operator on $\mathcal{H}$.
- $\|Ph\| \leq \|h\|$ for all $h \in \mathcal{H}$.
- $P^2 = P$ (P is idempotent)
- $\ker P = M^\perp$ and $\text{ran} P = M$.

- Conway's notation: $M \leq \mathcal{H}$ means M is a closed linear subspace of $\mathcal{H}$.
- Corollary: orthogonal decomposition of the space:
- Assume $M \leq \mathcal{H}$. For any $h \in \mathcal{H}$ we have $h = h_1 + h_\perp$ where $h_1 \in M$ and $h_\perp \in M^\perp$, and the decomposition is unique.

Corollary 2

If $M \leq \mathcal{H}$, then $(M^\perp)^\perp = M$.

Corollary 3

A linear subspace $A \subset \mathcal{H}$ is dense in $\mathcal{H}$ iff $A^\perp = \{0\}$.

Theorem 26 (Riesz Representation)

$L : \mathcal{H} \rightarrow \mathbb{R}$ bounded linear functional; there exists a unique $h_0 \in \mathcal{H}$ such that

$$L(h) = \langle h, h_0 \rangle, \quad \forall h \in \mathcal{H}.$$

Moreover, $\|L\| = \|h_0\|$.

- $\ker L$ closed linear subspace (since L is continuous).
- Assume $\ker L \neq \mathcal{H}$, hence $(\ker L)^\perp \neq \{0\}$;
- There exists $f_0 \in (\ker L)^\perp$ with $L(f_0) = 1$.
- Pick $h \in \mathcal{H}$ and let $\alpha = L(h)$.
- $L(h - \alpha f_0) = 0 \implies h - \alpha f_0 \in \ker L$, hence

$$0 = \langle h - \alpha f_0, f_0 \rangle = \langle h, f_0 \rangle - L(h)\|f_0\|^2$$

- $h_0 = \|f_0\|^{-2} f_0$ is the desired representer.
- Argue uniqueness as an exercise.

Definition 2

Consider a Hilbert space $\mathcal{H}$ of (real-valued) functions $f : \mathcal{X} \rightarrow \mathbb{R}$:

- $\mathcal{H}$ is a **reproducing kernel Hilbert space (RKHS)** if the evaluation functional $\delta_x : \mathcal{H} \rightarrow \mathbb{R}$ defined by

$$\delta_x(f) = f(x), \quad f \in \mathcal{H}$$

is continuous for all $x \in \mathcal{X}$.

- $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ a **reproducing kernel** of $\mathcal{H}$ if $K(\cdot, x) \in \mathcal{H}$ for all $x \in \mathcal{X}$ and the **reproducing property** holds

$$\langle f, K(\cdot, x) \rangle = f(x), \quad x \in \mathcal{X}, f \in \mathcal{H}$$

- In a RKHS, convergence in $\|\cdot\|_{\mathcal{H}}$ implies pointwise convergence, due to the continuity of evaluation functionals:

$$f_n \xrightarrow{\mathcal{H}} f \quad \implies \quad f_n(x) \rightarrow f(x), \quad \forall x \in \mathcal{X}$$

(LHS implies $\delta_x(f_n) \rightarrow \delta_x(f)$.)

- $L^2_{\text{pre}}([0, 1])$ can not be made into RKHS; $f_n(x) = x^n$, $f_n \xrightarrow{L^2} 0$ but $f_n(1) = 1$.
($L^2_{\text{pre}}([0, 1]) = C([0, 1])$ equipped with L^2 norm $(\int_0^1 f^2(t)dt)^{1/2}$)

Lemma 3

$\mathcal{H}$ is Hilbert function space over $\mathcal{X}$ that has reproducing kernel K . Then,

- K is a kernel (i.e., PSD).
- $\mathcal{H}$ is a RKHS.
- $K(\cdot, x)$ acts as the representer of evaluation functional δ_x ,
- as well as a feature map for K .

Definition 3

A symmetric bivariate function $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is called a kernel if it is PSD.

- K is PSD if $(K(x_i, x_j)) \in \mathbb{R}^{n \times n}$ is PSD for all $x_1, \dots, x_n \in \mathcal{X}$ and all $n \geq 1$.
- Equivalent to: $\exists \Phi : \mathcal{X} \rightarrow \mathcal{H}'$ such that

$$K(x, y) = \langle \Phi(x), \Phi(y) \rangle_{\mathcal{H}'}$$

- Φ is called a **feature map** of K .

Proof of Lemma 3

- We have $\delta_x(f) = f(x) = \langle f, K(\cdot, x) \rangle_{\mathcal{H}}$, hence (Cauchy-Schwarz)

$$|\delta_x(f)| \leq \|f\|_{\mathcal{H}} \|K(\cdot, x)\|_{\mathcal{H}}$$

showing that δ_x is bounded and $\|\delta_x\| \leq \|K(\cdot, x)\|_{\mathcal{H}}$; thus $\mathcal{H}$ is a RKHS.

- We have, again by reproducing property:

$$\langle K(\cdot, x), K(\cdot, y) \rangle_{\mathcal{H}} = K(x, y)$$

thus taking $\Phi(x) = K(\cdot, x)$ shows that K is a kernel with feature map Φ .

Converse

- Every RKHS has a unique reproducing kernel.

Proposition 3

Let $\mathcal{H}$ be an RKHS over $\mathcal{X}$, and let k_x be the Riesz representer of evaluation functional δ_x . Then $\mathcal{H}$ has a unique reproducing kernel which is given by $K(x, y) = \langle k_x, k_y \rangle_{\mathcal{H}}$ as well as

$$K(x, y) = \sum_{i \in I} e_i(x) e_i(y)$$

for any orthonormal basis $(e_i)_{i \in I}$ of $\mathcal{H}$.

Proof of Proposition 3

- Let $k_x \in \mathcal{H}$ be the representer of evaluation at x : $f(x) = \langle f, k_x \rangle_{\mathcal{H}}$.
- Define $K(x, y) := \langle k_x, k_y \rangle_{\mathcal{H}} = k_y(x) = k_x(y)$.
- That is $K(\cdot, y) = k_y$, hence

$$\langle K(\cdot, y), f \rangle_{\mathcal{H}} = \langle k_y, f \rangle_{\mathcal{H}} = f(y)$$

i.e., K is a reproducing kernel for $\mathcal{H}$.

- Let $\tilde{K}$ be a reproducing kernel of $\mathcal{H}$ and $(e_i)_{i \in I}$ an ONB. Then,

$$\tilde{K}(\cdot, y) = \sum_{i \in I} \langle \tilde{K}(\cdot, y), e_i \rangle e_i = \sum_{i \in I} e_i(y) e_i, \quad \forall y \in \mathcal{X}$$

where the series converges in $\|\cdot\|_{\mathcal{H}}$. Since $\mathcal{H}$ is a RKHS, the series also converges pointwise,

- i.e., $\tilde{K}(x, y) = \sum_{i \in I} e_i(y) e_i(x)$ for all $x, y \in \mathcal{X}$.
- Since $\tilde{K}$ was arbitrary, it should be unique.

Some examples

- $\mathcal{X} = \mathbb{R}^d$.
- Linear kernel $K(x, y) = \langle x, y \rangle$.
- Homogeneous polynomial kernel: $K(x, y) = \langle x, y \rangle^m$.
- E.g. for $m = 2$, $\Phi(x) = (x_i^2, \sqrt{2}x_i x_j : 1 \leq i < j \leq d)$ is a feature map.
- Inhomogeneous polynomial kernel: $K(x, y) = (1 + \langle x, y \rangle)^m$.
- Gaussian kernel: $K(x, y) = \exp(-\frac{1}{2\sigma^2} \|x - y\|^2)$. (Exercise: show it is PSD.)

Lemma 4

Let $\mathcal{H}$ be a RKHS on $\mathcal{X}$. Then, the linear span of $k_x(\cdot) = K(\cdot, x)$ is dense in $\mathcal{H}$.

- Proof: $\{k_x : x \in \mathcal{X}\}^\perp = \{0\}$
- Another way to state the lemma: $\overline{\text{span}\{k_x : x \in \mathcal{X}\}} = \mathcal{H}$.
- Note: the closure is w.r.t. the norm of $\mathcal{H}$.

□.

From RKHS to kernels

- The mapping from RKHSs to kernels is injective:

Proposition 4

Let $\mathcal{H}_i$ be a RKHS on $\mathcal{X}$ with kernel $K_i(x, y)$ for $i = 1, 2$. Then,

$$K_1(x, y) = K_2(x, y), \forall x, y \in \mathcal{X} \implies \mathcal{H}_1 = \mathcal{H}_2, \|f\|_{\mathcal{H}_2} = \|f\|_{\mathcal{H}_1}$$

- $k_x := K_1(\cdot, x) = K_2(\cdot, x)$.
- $W = \text{span}\{k_x : x \in \mathcal{X}\} \implies W$ is dense in $\mathcal{H}_i, i = 1, 2$.
- Easy to verify that $\|f\|_{\mathcal{H}_1} = \|f\|_{\mathcal{H}_2}$ for $f \in W$.
- If $f \in \mathcal{H}_1$, then $\exists \{f_n\} \subset W, f_n \xrightarrow{\mathcal{H}_1} f$.
- $\{f_n\}$ is Cauchy in $(W, \|\cdot\|_{\mathcal{H}_1})$, hence Cauchy in $(W, \|\cdot\|_{\mathcal{H}_2})$,
- hence $\exists g \in \mathcal{H}_2$ s.t. $f_n \xrightarrow{\mathcal{H}_2} g$.
- $f_n(x) \rightarrow g(x)$ pointwise $\forall x$, and also $f_n(x) \rightarrow f(x)$ pointwise for all x ,
- hence $f(x) = g(x)$, i.e., $f \in \mathcal{H}_2$. (Reverse argument holds as well.)
- This proves $\mathcal{H}_1 = \mathcal{H}_2$. Since the norms are equal over a dense subset, they are equal everywhere.

From kernels to RKHS

- Every kernel has a (unique) RKHS.

Theorem 27 (Moore or Moore-Aronszajn)

Let $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a (PSD) kernel. There exists a RKHS (of functions) on $\mathcal{X}$ with reproducing kernel K .

- $W = \text{span}\{k_x : x \in \mathcal{X}\}$. Define an inner product on W by

$$\left\langle \sum_i \alpha_i k_{x_i}, \sum_j \beta_j k_{x_j} \right\rangle_{\star} := \sum_{i,j} \alpha_i \beta_j K(x_i, x_j)$$

- Can verify that it is well-defined: same inner product for different representations of the same function as linear combinations of $k_x, x \in \mathcal{X}$. (Show that $f(x) = \sum_i \alpha_i k_{x_i}(x)$ is zero identically iff $\langle f, w \rangle_{\star} = 0, \forall w \in W$.)
- By definition for any $f \in W$, $\langle f, k_x \rangle_{\star} = f(x)$.
- W is an inner product space, hence has a completion (as equivalence classes of Cauchy sequences) which is a Hilbert space.
- Any element of the completion is in fact a function (as opposed to the completion of $C([0, 1])$ for example to get $L^2([0, 1])$.) See next slide.

- Let $h \in \mathcal{H}$, and $\{f_n\} \subset W$ a Cauchy seq. with $f_n \xrightarrow{\mathcal{H}} h$.
- By Cauchy–Schwarz inequality

$$|f_n(x) - f_m(x)| = \langle f_n - f_m, k_x \rangle_{\star} \leq \sqrt{K(x, x)} \|f_n - f_m\| \quad (19)$$

(valid by our definition of $\langle \cdot, \cdot \rangle_{\star}$ over W)

- Hence, f_n is pointwise convergent, and we can define h to be $h(x) = \lim_n f_n(x)$. Usual argument shows that this is independent of the particular Cauchy sequence⁵.
- Let $\langle \cdot, \cdot \rangle$ be the inner product on $\mathcal{H}$, and f_n and h as above. Then,

$$\langle h, k_x \rangle = \lim_n \langle f_n, k_x \rangle = \lim_n \langle f_n, k_x \rangle_{\star} = \lim_n f_n(x) = h(x)$$

i.e., k_x is the representer of evaluation in $\mathcal{H}$.

- Hence $\mathcal{H}$ is a RKHS, and $K(x, y) = k_x(y)$ is its unique reproducing kernel (see Lemma 3 and Proposition 3).

⁵Recall that two Cauchy sequences $\{f_n\}$ and $\{g_n\}$ are equivalent if $\lim_n \|f_n - g_n\|_{\star} = 0$.

- Conclusion: there is a one-to-one mapping between kernels and RKHSs.

Example 5 (Sobolev order 1)

- Consider the following:

$$H^1([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R} \mid f(0) = 0, \text{ and } f \text{ is abs. cont., } f' \in L^2([0, 1])\}$$

- With inner product

$$\langle f, g \rangle_{H^1} = \int_0^1 f'(t)g'(t)dt$$

- Verify that it is a Hilbert space.
- Take $k_t(s) := K(s, t) = \min\{s, t\}$ and note that $k'_t(s) = 1\{s < t\}$ a.e. s .
- Note that $k_t \in H^1$ and satisfies the reproducing property:

$$\int_0^1 f'(s)k'_t(s)ds = f(t), \quad \forall t \in [0, 1], f \in H^1$$

- It follows (Lemma 3) that $H^1([0, 1])$ is a RKHS with kernel K .
- Conclude that K is PSD, i.e., $(\min\{t_i, t_j\}) \in \mathbb{R}^{n \times n}$ is PSD for any $t_1, \dots, t_n \in [0, 1]$ which might not be easy to show directly.

Example 6 (Higher-order Sobolev, smoothing splines)

- $H^\alpha([0, 1]) = f$ such that $f^{(\alpha-1)}$ is abs. cont and $f^{(\alpha)} \in L^2([0, 1])$.
- Additional boundary condition $f(0) = f^{(1)}(0) = \dots = f^{(\alpha-1)}(0) = 0$.
- Inner product $\langle f, g \rangle_{H^\alpha} = \int_0^1 f^{(\alpha)}(t)g^{(\alpha)}(t)dt$.
- H^α is a RKHS with the following kernel:

$$k_x(y) = K(x, y) = \int_0^1 \psi_\alpha(x-t)\psi_\alpha(y-t)dt, \quad \psi_\alpha(t) = \frac{t_+^{\alpha-1}}{(\alpha-1)!}.$$

- $k_x \in H^\alpha$ and $k_x^{(\alpha)}(y) = \psi_\alpha(x-y)$. [Note: $\psi_\alpha^{(\alpha-1)}(t) = 1\{t > 0\}$]
- Reproducing property follows from Taylor expansion

$$f(x) = \sum_{\ell=1}^{\alpha-1} f^{(\ell)}(0) \frac{x^\ell}{\ell!} + \int_0^1 f^{(\alpha)}(t)\psi_\alpha(x-t)dt$$

Interpolation

- Consider the minimum norm interpolation

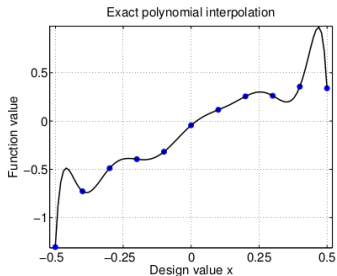
$$\min_{f \in \mathcal{H}} \|f\|_{\mathcal{H}} \quad f(x_i) = y_i, \quad i = 1, \dots, n. \quad (20)$$

- Let $\mathbf{K} = \frac{1}{n}(K(x_i, x_j)) \in \mathbb{R}^{n \times n}$ be the kernel matrix, and $y = (y_i) \in \mathbb{R}^n$.

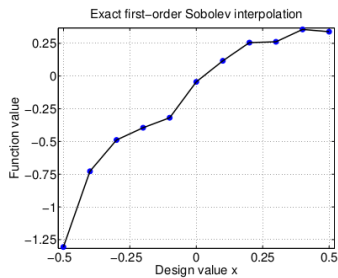
Proposition 5

Problem (20) is feasible iff $y \in \text{ran}(\mathbf{K})$. In that case the (unique) optimal solution is

$$\hat{f}(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \hat{\alpha}_i K(\cdot, x_i), \quad \mathbf{K} \hat{\alpha} = \frac{y}{\sqrt{n}}$$



(a)



(b)

Figure 12.1 Exact interpolation of $n = 11$ equally sampled function values using RKHS methods. (a) Polynomial kernel $\mathcal{K}(x, z) = (1 + xz)^{12}$. (b) First-order Sobolev kernel $\mathcal{K}(x, z) = 1 + \min\{x, z\}$.

- Let $L_x = \text{span}\{K(\cdot, x_i) : i = 1, \dots, n\} = \{f_\alpha : \alpha \in \mathbb{R}^n\}$ where

$$f_\alpha := \frac{1}{\sqrt{n}} \sum_{i=1}^n \alpha_i K(\cdot, x_i)$$

- L_x is a linear subspace of $\mathcal{H}$ with $\dim(L_x) \leq n$, hence closed.
- Minimum norm solution should be in L_x :
- Take $h \in \mathcal{H}$ and let $h = h_1 + h_\perp$ be its decomposition w.r.t. $L \oplus L^\perp$.
- $h_\perp(x_i) = 0$ for all i , so h is feasible iff h_1 is feasible.
- Assume h is feasible, hence h_1 is feasible.
- $\|h\|^2 = \|h_1\|^2 + \|h_\perp\|^2$. Hence, $h_\perp \neq 0 \implies \|h\| > \|h_1\|$.
- $\implies$ the minimum norm solution should have $h_\perp = 0$, i.e., $h \in L_x$.
- The problem reduces to

$$\begin{aligned} \min_{f \in L_x} \quad & \|f\|_{\mathcal{H}} \\ & f(x_i) = y_i, \forall i, \end{aligned} \quad \equiv \quad \begin{aligned} \min_{\alpha \in \mathbb{R}^n} \quad & \|f_\alpha\|_{\mathcal{H}} \\ & f_\alpha(x_i) = y_i, \forall i, \end{aligned} \quad \equiv \quad \begin{aligned} \min_{\alpha \in \mathbb{R}^n} \quad & \alpha^T K \alpha \\ & K \alpha = y / \sqrt{n} \end{aligned} \quad (21)$$

ERM with RKHS regularization

- Consider the following optimization

$$\min_{f \in \mathcal{H}} L(f(x_1), \dots, f(x_n)) + \omega(\|f\|_{\mathcal{H}}^2) \quad (22)$$

Theorem 28

Let H be an RKHS in X , $\omega : \mathbb{R} \rightarrow \mathbb{R}$ be increasing, and $L : \mathbb{R}^n \rightarrow \mathbb{R}$. Then, any minimizer f^* of (22) is in the span of $\{k_{x_1}, \dots, k_{x_n}\}$.

- Let us write $\ell_x(f) = L(f(x_1), \dots, f(x_n))$.
- Let $L_x = \text{span}\{K(\cdot, x_i) : i = 1, \dots, n\}$,
- Let f be optimal and write $f = f_1 + f_{\perp}$ where $f_1 \in L_x$ and $f_{\perp} \perp L_x$.
- $f_{\perp}(x_i) = 0$ for all i , hence $\ell_x(f) = \ell_x(f_1)$.
- $\omega(\|f\|_{\mathcal{H}}^2) = \omega(\|f_1\|_{\mathcal{H}}^2 + \|f_{\perp}\|^2) \geq \omega(\|f_1\|_{\mathcal{H}}^2)$.
- Strict inequality unless $\|f_{\perp}\|_{\mathcal{H}} = 0 \implies f_{\perp} = 0$.

ERM with RKHS regularization

- Example: Regularized ERM

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \lambda \|f\|_{\mathcal{H}}^2 \quad (23)$$

- We can restrict to solutions of the form

$$f_{\alpha} := \frac{1}{\sqrt{n}} \sum_{i=1}^n \alpha_i K(\cdot, x_i)$$

- We have $\|f_{\alpha}\|_{\mathcal{H}}^2 = \alpha^T \mathbf{K} \alpha$, and

$$f_{\alpha}(x_j) = \langle K(\cdot, x_j), f_{\alpha} \rangle_{\mathcal{H}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \alpha_i K(x_j, x_i) = \sqrt{n} \mathbf{K} \alpha$$

- Problem reduces to

$$\min_{\alpha \in \mathbb{R}^n} \frac{1}{n} \sum_{i=1}^n L(y_i, \sqrt{n}(\mathbf{K} \alpha)_i) + \lambda \alpha^T \mathbf{K} \alpha \quad (24)$$

Special cases

- Kernel ridge regression (KRR):

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 \quad (25)$$

- Can be thought of as solving the noisy interpolation problem:
 $y_i = f^*(x_i) + w_i$.
- Solution will be $f_{\hat{\alpha}}$ with

$$\hat{\alpha} = (\mathbf{K} + \lambda I_n)^{-1} \frac{\mathbf{y}}{\sqrt{n}}$$

- Kernel SVM:

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (1 - y_i f(x_i))_+ + \lambda \|f\|_{\mathcal{H}}^2 \quad (26)$$

- Derive the dual as an exercise.

Mercer Theorem

Definition 4

Let $\mathcal{X} \subset \mathbb{R}^d$ be compact. $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is called a Mercer kernel if it is continuous and PSD, i.e., a continuous kernel function.

Lemma 5

Let K be a Mercer kernel, then $\mathcal{H}(K)$ is separable.

- Since $\mathcal{H}(K)$ is separable (has a countable dense subset) it has a countable orthonormal basis (ONB).

Lemma 6

Let K be a Mercer kernel and let $\{e_n : n \in \mathbb{N}\}$ be an ONB for $\mathcal{H}(K)$, with $K(x, y) = \sum_{n=1}^{\infty} e_n(x)e_n(y)$. Then the series converges absolutely and uniformly.

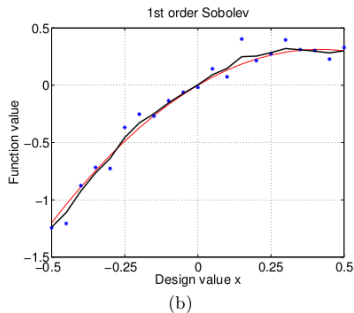
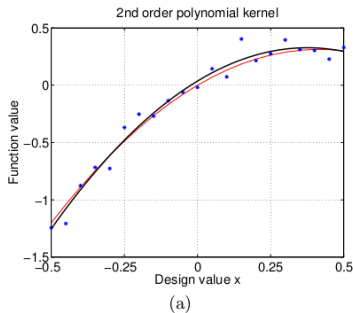


Figure 12.2 Illustration of kernel ridge regression estimates of function $f^*(x) = \frac{3x}{2} - \frac{9}{5}x^2$ based on $n = 21$ samples, located at design points $x_i = -0.5 + 0.05(i - 1)$ over the interval $[-0.5, 0.5]$. (a) Kernel ridge regression estimate using the second-order polynomial kernel $\mathcal{K}(x, z) = (1 + xz)^2$ and regularization parameter $\lambda_n = 0.01$. (b) Kernel ridge regression estimate using the first-order Sobolev kernel $\mathcal{K}(x, z) = 1 + \min\{x, z\}$ and regularization parameter $\lambda_n = 0.08$.

- The integral operator associated with K assuming $\mathcal{X}$ is equipped with a measure μ ,

$$(T_K f)(x) = \int_{\mathcal{X}} K(x, y) f(y) d\mu(y)$$

- If $K \in L^2(\mathcal{X} \times \mathcal{X}, \mu \otimes \mu)$, then T_K is well-defined for all $f \in L^2(\mathcal{X}, \mu)$.
- Think of it as a continuous version of matrix multiplication.

Theorem 29

Let $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a Mercer kernel with μ a finite Borel measure on $\mathcal{X}$. Then, the integral operator T_K is a bounded map from $L^2(\mathcal{X}, \mu)$ into itself with

$$\|T_K\| \leq \left(\int_{\mathcal{X}} \int_{\mathcal{X}} K(x, y)^2 d\mu(x) d\mu(y) \right)^{1/2}$$

and every function in the range of T_K is continuous.

- T_K will be in fact a Hilbert-Schmidt operator.

Theorem 30 (Mercer)

- Let K be a Mercer kernel on $\mathcal{X}$,
- μ finite Borel measure with support $\mathcal{X}$, and
- $T_K : L^2(\mathcal{X}, \mu) \rightarrow L^2(\mathcal{X}, \mu)$ its associated integral operator.
- Then, there is a countable collection of continuous function $\{e_n\}$ on $\mathcal{X}$ **orthonormal in L^2**
- that are eigenvectors of T_K with corresponding eigenvalues $\{\lambda_n\}$
- such that for every $f \in L^2(X, \mu)$,

$$T_K f = \sum_i \lambda_i \langle f, e_i \rangle e_i.$$

- Furthermore, $K(x, y) = \sum_i \lambda_i e_i(x) e_i(y)$.
(with the convergence uniform and absolute).
- It follows that $\{e_i, i \in \mathbb{N}\}$ is an **orthogonal basis** for $\mathcal{H}(K)$.
- $\{e_i\}$ are not normalized in $\mathcal{H}(K)$.
- Can guess that $\{\sqrt{\lambda_i} e_i\}$ is an orthonormal basis for $\mathcal{H}(K)$.

Mercer representation of the RKHS

Proposition 6

Under the Mercer theorem assumptions:

$$\mathcal{H}(K) = \left\{ \sum_i a_i \sqrt{\lambda_i} e_i : (a_i) \in \ell_2 \right\}.$$

If $f = \sum_i a_i \sqrt{\lambda_i} e_i$ and $g = \sum_i b_i \sqrt{\lambda_i} e_i$, then $\langle f, g \rangle_{\mathcal{H}} = \sum_i a_i b_i$.

Operations on kernels

Theorem 31 (Aronszajn's inclusion)

Let $\mathcal{X}$ be a set, and $K_i : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}, i = 1, 2$ kernels on $\mathcal{X}$. Then,

$$\mathcal{H}(K_1) \subseteq \mathcal{H}(K_2) \iff \exists c > 0, K_1 \leq c^2 K_2$$

Moreover $\|f\|_2 \leq c\|f\|_1$ for all $f \in \mathcal{H}(K_1)$.

- $K_1 \leq c^2 K_2$ should be interpreted as $c^2 K_2 - K_1$ is PSD.

Theorem 32 (Aronszajn's sums of kernels)

Let $\mathcal{H}_i$ be a RKHS with kernel K_i and norm $\|\cdot\|_i$, for $i = 1, 2$.

Let $K = K_1 + K_2$. Then,

$$\mathcal{H}(K) = \{f_1 + f_2 : f_i \in \mathcal{H}_i, i = 1, 2\}$$

with norm $\|\cdot\|$ given by

$$\|f\|^2 = \min\{\|f_1\|_1^2 + \|f_2\|_2^2 : f = f_1 + f_2, f_i \in \mathcal{H}_i, i = 1, 2\}$$

- If $\mathcal{H}_1 \cap \mathcal{H}_2 = \{0\}$, there is a unique representation $f = f_1 + f_2$, hence

$$\|f\|_{\mathcal{H}(K)}^2 = \|f_1\|_1^2 + \|f_2\|_2^2$$

Example 7

- Recall the 1st order Sobolev kernel $K_2(s, t) = \min\{s, t\}$ with RKHS

$$\mathcal{H}_2 = \{f : [0, 1] \rightarrow \mathbb{R} \mid f(0) = 0, \text{ and } f \text{ is abs. cont., } f' \in L^2([0, 1])\}$$

- Let $\mathcal{H}_1 = \text{span}(\{1\})$, the space of constant functions.
- $\mathcal{H}_1$ is a RKHS: $\langle f, g \rangle_1 = f(0)g(0)$ and $K_1(s, t) = 1$.
- Note $\mathcal{H}_1 \cap \mathcal{H}_2 = \{0\}$.
- Hence, $K(s, t) = 1 + \min\{s, t\}$ generates the RKHS

$$\begin{aligned}\mathcal{H} &= \{f : [0, 1] \rightarrow \mathbb{R} : f = c1 + f_2, f_2 \in \mathcal{H}_2, c \in \mathbb{R}\} \\ &= \{f : [0, 1] \rightarrow \mathbb{R} : f \text{ is abs. cont., } f' \in L^2([0, 1])\}\end{aligned}$$

with norm

$$\|f\|_{\mathcal{H}}^2 = f^2(0) + \int_0^1 [f'(t)]^2 dt$$

- Pull-back of a kernel K by map ϕ is

$$K \circ \phi(s, t) = K(\phi(s), \phi(t))$$

which itself is a kernel.

Theorem 33 (Pull-back)

- Let $\mathcal{X}$ and $\mathcal{S}$ be sets.
- Let $\phi : \mathcal{S} \rightarrow \mathcal{X}$ be a function and $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a kernel.
- Then, $\mathcal{H}(K \circ \phi) = \{f \circ \phi : f \in \mathcal{H}(K)\}$ and

$$\|u\|_{\mathcal{H}(K \circ \phi)} = \min\{\|f\|_{\mathcal{H}(K)} : u = f \circ \phi\}$$

Other results:

- Tensor product of kernel

$$K((x, s), (y, t)) = K_1(x, y)K_2(s, t)$$

denoted as $K_1 \otimes K_2$.

- Product of kernels

$$K(x, y) = K_1(x, y)K_2(x, y)$$

denoted as $K_1 \odot K_2$.

- Letting $\Delta(x) = (x, x)$ be the diagonal map, we have

$$K_1 \odot K_2 = (K_1 \otimes K_2) \circ \Delta$$

Graphical models

- Graphical models: encoding conditional independence assumptions using graph separation.
- Different kind of graphs (and graph separations) encode different kinds of conditional independence assumptions.
- Two common categories are directed and undirected graphical models.
- Directed case often called a Bayesian network, used to model causal relations.

Properties of conditional independence

Symmetry: $X \perp Y \mid Z \implies Y \perp X \mid Z$.

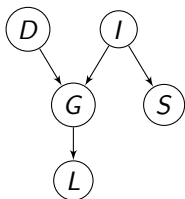
Decomposition: $X \perp (Y, W) \mid Z \implies X \perp Y \mid Z$.

Weak union: $X \perp (Y, W) \mid Z \implies X \perp Y \mid Z, W$.

The following only holds for strictly positive distributions:

Intersection: $X \perp Y \mid (W, Z)$, and $X \perp W \mid (Y, Z) \implies X \perp (Y, W) \mid Z$

Directed graphical models



- D: Difficulty (Stat 100C)
- I: Intelligence
- G: Grade (Stat 100C)
- L: (Recom.) Letter
- S: SAT Score

Definition 5 (BN structure)

- A Bayesian network structure/graph is a directed acyclic graph (DAG) G .
- Nodes represent random variables $X_1, \dots, X_n$.
- $\text{Pa}_{X_i}^G$: parents of X_i in G .
- $\text{Nd}_{X_i}^G$: non-descendants of X_i in G .
- G encodes the following set of conditional independence assumptions:

$$X_i \perp \text{Nd}_{X_i}^G \mid \text{Pa}_{X_i}^G, \quad \forall i,$$

called **local independences** denoted by $\mathcal{I}_\ell(G)$.

- Also called directed local Markov property.

Definition 6

For a distribution P , let $\mathcal{I}(P)$ be the set of (conditional) independence assertions of the form $(X \perp Y \mid Z)$ that hold in P .

Definition 7

We say that G is an **I-map** for P whenever $\mathcal{I}_\ell(G) \subseteq \mathcal{I}(P)$.

- I.e, G is an I -map for P whenever P is locally Markov w.r.t. G .

Definition 8 (Factorization)

- Let G be a BN graph over variables $X = (X_1, \dots, X_n)$.
- We say that distribution P factorizes according to G if P can be expressed as

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}_{X_i}^G)$$

- Chain rule of BNs.
- $P(X_i \mid \text{Pa}_{X_i}^G)$ local probabilistic models.

Definition 9

A Bayesian network is a pair (G, P) where P factorizes over G .

I-map to factorization

Theorem 34

G a BN structure over X , and P a joint distribution for X :

G is an I-map for $P \implies P$ factorizes according to G

- Assume $X_1, \dots, X_n$ are in a topological ordering.
- $\{X_1, \dots, X_{i-1}\} = \text{Pa}_{X_i} \cup \mathcal{Z}$ where $\mathcal{Z} \subseteq \text{Nd}_{X_i}$.
- By assumption $X_i \perp \text{Nd}_{X_i} \mid \text{Pa}_{X_i}$, hence $X_i \perp \mathcal{Z} \mid \text{Pa}_{X_i}$
- That is, $P(X_i \mid X_1, \dots, X_{i-1}) = P(X_i \mid \text{Pa}_{X_i})$, hence

$$p(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid X_1, \dots, X_{i-1}) = \prod_{i=1}^n P(X_i \mid \text{Pa}_{X_i}).$$

Factorization to I -map

Theorem 35

G a BN structure over X , and P a joint distribution for X :

$$P \text{ factorizes according to } G \implies G \text{ is an } I\text{-map for } P$$

Conclusion:

- P factorizes over G iff G is an I -map for P . In other words,

$$P \text{ factorizes according to } G \iff \mathcal{I}_\ell(G) \subseteq \mathcal{I}(P)$$

d -separation

- Are there other independencies that can be “read from the G ”, besides the local ones?
- Suppose that P factorizes according to G ; then $I_\ell(G) \subseteq \mathcal{I}(P)$.
- Is there a larger set $\mathcal{I}(G)$ such that

$$\mathcal{I}_\ell(G) \subseteq \mathcal{I}(P) \implies \mathcal{I}_\ell(G) \subseteq \mathcal{I}(G) \subseteq \mathcal{I}(P)$$

- Yes.
- Need to characterize information flow over the network.

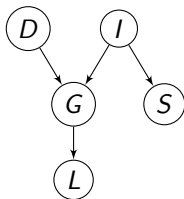
- When can we have $X \perp Y \mid Z$?
- Consider the following four cases:
 - Causal trail $X \rightarrow Z \rightarrow Y$
 - Evidential trail $X \leftarrow Z \leftarrow Y$
 - Common cause $X \leftarrow Z \rightarrow Y$
 - Common effect $X \rightarrow Z \leftarrow Y$ a.k.a **v-structure** or **collider**
- The trail is active if information flows through it, otherwise blocked.
- The first three are blocked iff Z is observed.
- v-structure is active iff either Z or one of its descendants is observed.

Definition 10

- G a BN structure
- $X_1 \rightleftharpoons X_2 \rightleftharpoons \cdots \rightleftharpoons X_n$ a trail in G
- Z a subset of nodes.

Trail $X_1 \rightleftharpoons X_2 \rightleftharpoons \cdots \rightleftharpoons X_n$ is active given Z if

- For every $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$, either X_i or one of its descendants are in Z ,
- no other node along the trail is in Z



- Trail $D \rightarrow G \leftarrow I \rightarrow S$.
- For $Z = \emptyset$, not active, since v -structure $D \rightarrow G \leftarrow I$ is not active.
- For $Z = \{L\}$, active, since v -structure is active, and no other block.
- For $Z = \{L, I\}$, not active, observing I block $G \leftarrow I \rightarrow S$.

Definition 11

- X, Y, Z there set of variables.
- X, Y are d-separated given Z , denoted $\text{d-sep}_G(X; Y \mid Z)$ if
- given Z , there is no active trail with endpoints in X and Y . Define

$$\mathcal{I}(G) := \{(X \perp Y \mid Z) : \text{d-sep}_G(X; Y \mid Z)\}$$

- $\mathcal{I}_G$ can be thought of as the set of directed global Markov properties.

Theorem 36

If P factorizes according to G , then $\mathcal{I}(G) \subseteq \mathcal{I}(P)$.

- Equivalent statement $\mathcal{I}_\ell(G) \subset \mathcal{I}(P) \implies \mathcal{I}(G) \subset \mathcal{I}(P)$.
- (Reverse direction is true and clear since $\mathcal{I}_\ell(G) \subseteq \mathcal{I}(G)$. Why?)

Definition 12

A distribution P is **faithful** to G whenever

$$(X \perp Y \mid Z) \in \mathcal{I}(P) \implies \text{d-sep}_G(X; Y \mid Z),$$

that is, $\mathcal{I}(P) \subseteq \mathcal{I}(G)$.

- Conclusion:
If P factorizes over G , **and** P is faithful to G , then $\mathcal{I}(P) = \mathcal{I}(G)$.
- Not all distributions that factorize over G are faithful to G .
- Any product measure P on X (i.e., all variables are independent unconditionally) factorizes over any graph G , but is faithful only the empty graph.

Definition 13 (P-map)

G is a **perfect** map (P-map) for P whenever $\mathcal{I}(P) = \mathcal{I}(G)$.

- Is there anything even bigger than $\mathcal{I}(G)$ contained in every $\mathcal{I}(P)$ for P factorizing according to G ?
- No.

Theorem 37

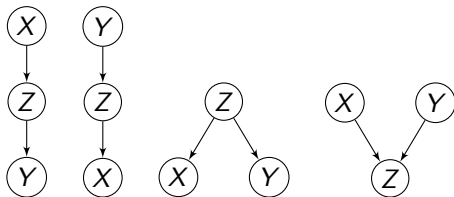
Let G be a BN structure. If X and Y are not d -separated in G given Z , then they are dependent given Z in some distribution that factorizes over G .

- Equivalently

$$\bigcap_{P \text{ factorizes over } G} \mathcal{I}(P) = \mathcal{I}(G)$$

I-Equivalence

- Can we have $G_1 \neq G_2$ but $\mathcal{I}(G_1) = \mathcal{I}(G_2)$? Yes.
- G_1 and G_2 are called *I*-equivalent whenever $\mathcal{I}(G_1) = \mathcal{I}(G_2)$.
- Also called Markov equivalent.
- A proper equivalence relation;
- partitions the set of DAGs (into Markov equivalence classes).
- Among the four 3-DAGs, the first three are equivalent:



- $X \rightarrow Y$ and $Y \leftarrow X$ are equivalent. (Encode no independence.)

Definition 14

Skeleton of a DAG is the undirected graph obtained by removing edge directions.

Theorem 38

If G_1 and G_2 have the same skeleton and the same v -structures, then G_1 and G_2 are equivalent.

- Same skeleton and v -structures is sufficient but not necessary.
- Consider complete DAGs (i.e., acyclic/transitive tournaments), encode no independence, v -structures are different.

Definition 15

A v -structure $X \rightarrow Z \leftarrow Y$ is an immorality if there is no edge between X and Y .

Theorem 39

G_1 and G_2 are I -equivalent iff they have the same skeleton and immoralities.

- Complete DAGs have no immoralities (but possibly many v -structures.)

Definition 16

An edge $X \rightarrow Y$ in G is covered if $\text{Pa}_Y^G = \text{Pa}_X^T \cup \{X\}$.

Theorem 40

G and G' are I-equivalent iff $\exists$ a sequence of networks $G = G_1, \dots, G_k = G'$, all I-equivalent to G such that G_{i+1} differs from G_i only in reversal of a single covered edge.

Definition 17

G is a minimal I -map for P if G is an I -map for P (i.e., $\mathcal{I}(G) \subseteq \mathcal{I}(P)$) and removing any edge from G renders it not an I -map.

- A complete DAG K is an I -map for a any P (since $\mathcal{I}(K) = \emptyset$) but often not a minimal I -map.

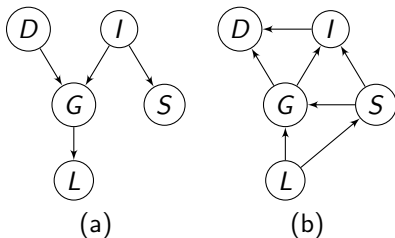
How to get a minimal I -map

- Fix an ordering.
- For every i , choose $U \subseteq X_1^{i-1} := \{X_1, \dots, X_{i-1}\}$ to be the minimal subset satisfying

$$X_i \perp (X_1^{i-1} \setminus U) \mid U,$$

minimal means removing any element to U violates the property.

- U will be Pa_{X_i} .
- There could be multiple choices. Unique if P is positive.
- Every ordering could potentially lead to a different minimal I -MAP (even if P is positive).



- (a) corresponds to the ordering D, I, G, S, L . (Multiple other topological orderings)
- Suppose that P factors over G and is faithful to G .
- I.e., G is a P-map for P , or equivalently $\mathcal{I}(G_a) = \mathcal{I}(P)$.
- Then (b) is the minimal I-map associated with the ordering L, S, G, I, D .
- $\mathcal{I}(G_b) \subset \mathcal{I}(P)$, but $\mathcal{I}(G_b) \neq \mathcal{I}(P)$.
- Easiest to see by noting that G_a and G_b are not Markov equivalent,
- $\mathcal{I}(G_a) \neq \mathcal{I}(G_b)$ since they have different skeletons.

Definition 18

A Gibbs distributions over $X_1, \dots, X_n$ with factors $\Phi = (\phi_i(X_{S_i}))$ where $S_i \subset [n]$ is

$$P_\Phi(X_1, \dots, X_n) = \frac{1}{Z} \prod_i \phi_i(X_{S_i}).$$

- Example: $P_\Phi(X_1, X_2, X_3) = \frac{1}{Z} \phi(X_1, X_2) \phi_2(X_2, X_3)$.

Definition 19 (Markov network)

P_Φ with factors $\Phi = (\phi_1(X_{S_1}), \dots, \phi_k(X_{S_k}))$, factorizes over an undirected graph G , if each of $S_1, \dots, S_k$ is a clique of G .

- Clique = complete (induced) subgraph.
- P_Φ is Markov w.r.t. G , or G is a Markov network for P_Φ .
- $\phi_i(S_i)$ is called clique potential.
- Can reduce to maximal clique. (Possible loss of information.)

Markov properties

Definition 20

Let G be a Markov network (undirected graph).

A path $X_1 - X_2 - \dots - X_n$ in G is active given Z , if none of X_i is in Z .

Definition 21

Z separates X and Y in G , denoted $\text{sep}_H(X; Y|Z)$ if there is no active path given Z with endpoints in X and Y .

Global independences associated with G is defined as

$$\mathcal{I}(G) = \{(X \perp Y \mid Z) : \text{sep}_G(X; Y \mid Z)\}$$

- Usual notion of separation, every path between X and Y is block by some node in Z .
- $\mathcal{I}(G)$ is the set of global Markov properties/independencies.

Theorem 41

Let G be an undirected graph (Markov network). Then,

$$P \text{ factorizes over } G \implies G \text{ is an I-map for } P$$

- The converse is not always true.

Theorem 42 (Hammersley-Clifford)

Let G be an undirected graph (Markov network). If P is a *positive distribution*,

$$G \text{ is an I-map for } P \implies P \text{ factorizes over } G$$

- Assume that Z separates Y_1 and Y_2 .
- Also assume that Y_1, Y_2, Z partition the whole set of variables $\mathcal{X}$.
- I.e., $Y_1 \cup Y_2 \cup Z = \mathcal{X}$.
- Every clique of G is either fully in $Y_1 \cup Z$ or $Y_2 \cup Z$, hence

$$P(X_1, \dots, X_n) \propto \prod_{i \in I_1} \phi_i(S_i) \prod_{i \in I_2} \phi_i(S_i) = f(Y_1, Z) f(Y_2, Z)$$

where $S_i \subseteq Y_r \cup Z$ for $i \in I_r$ and $r = 1, 2$.

- This proves $Y_1 \perp Y_2 \mid Z$ under P .
- In general let $U = \mathcal{X} \setminus (Y_1 \cup Y_2 \cup Z)$.
- Then, U can be partitioned into $U = U_1 \cup U_2$ such that
- Z separates $Y_1 \cup U_1$ from $Y_2 \cup U_2$.
- It follows that $(Y_1 \cup U_1) \perp (Y_2 \cup U_2) \mid Z$,
- hence $Y_1 \perp Y_2 \mid Z$.

Local Markov properties

- Let $\mathcal{X} = \{X_1, \dots, X_n\}$.
- Pairwise Markov properties associated with G :

$$\mathcal{I}_p(G) = \left\{ (X \perp Y \mid \mathcal{X} \setminus \{X, Y\}) : X - Y \notin G \right\}$$

- Local Markov properties associated with G :

$$\mathcal{I}_\ell(G) = \left\{ (X \perp \mathcal{X} \setminus \text{cl}_G(X) \mid \text{bd}_G(X)) : X \in \mathcal{X} \right\}$$

- $\text{bd}(X)$ = boundary of X = Markov blanket of X = neighbors of X .
- $\text{cl}(X) = \{X\} \cup \text{bd}(X)$.
- For any distribution: pairwise $\implies$ local $\implies$ global.
- The three are equivalent for positive distributions.